

## THE TABLEAU METHOD

We will shortly be studying *first-order logic*, which, as we shall see, is an enormous advance over propositional logic and which, in fact, is an adequate logical framework for the entire body of mathematics! For this vital subject, the method of truth tables will not at all suffice, so we now turn to the method known as *tableau*, which in this chapter will be treated on the elementary propositional level, and which will be extended to first-order logic in a later chapter.

We are now dealing with *formulas*, as defined in Chapter 7. We are defining an *interpretation* of a formula to be a specification as to which of its propositional variables are to be interpreted as *true* and which as *false*. For a formula with only one propositional variable  $p$ , there are two and only two interpretations: (1)  $p$  true; (2)  $p$  false. For a formula with two propositional variables—say,  $p$  and  $q$ —there are four interpretations: (1)  $p$  true,  $q$  true; (2)  $p$  true,  $q$  false; (3)  $p$  false,  $q$  true; (4)  $p$  false,  $q$  false. With three propositional variables, there are eight interpretations, and, in general, for  $n$  variables there are  $2^n$  possible interpretations. A truth table for a formula  $X$  is nothing more than a systematic device for determining under which interpretations  $X$  is true and under which  $X$  is false—each line in the truth table corresponds to one of the interpretations. We recall that a formula is called a *tautology* if it is true under all interpretations, and it is the tautologies that are of particular interest.

## THE METHOD OF TABLEAUX

We begin by noting that under any interpretation the following eight facts hold (for any formulas  $X, Y$ ):

- (1) • If  $\sim X$  is true, then  $X$  is false.  
• If  $\sim X$  is false, then  $X$  is true.
- (2) • If a conjunction  $X \wedge Y$  is true, then  $X, Y$  are both true.  
• If a conjunction  $X \wedge Y$  is false, then either  $X$  is false or  $Y$  is false.
- (3) • If a disjunction  $X \vee Y$  is true, then either  $X$  is true or  $Y$  is true.  
• If a disjunction  $X \vee Y$  is false, then  $X, Y$  are both false.
- (4) • If  $X \Rightarrow Y$  is true, then either  $X$  is false or  $Y$  is true.  
• If  $X \Rightarrow Y$  is false, then  $X$  is true and  $Y$  is false.

These eight facts provide the basis for the tableau method.

### SIGNED FORMULAS

At this stage it will prove useful to introduce the symbols  $T$  and  $F$  to our object language and define a *signed formula* as an expression  $TX$  or  $FX$ , where  $X$  is an unsigned formula. (Informally, we read " $TX$ " as " $X$  is true" and " $FX$ " as " $X$  is false.")

DEFINITION. Under any interpretation, a signed formula  $TX$  is called *true* if  $X$  is true and *false* if  $X$  is false. And a signed formula  $FX$  is called *true* if  $X$  is false and *false* if  $X$  is true.

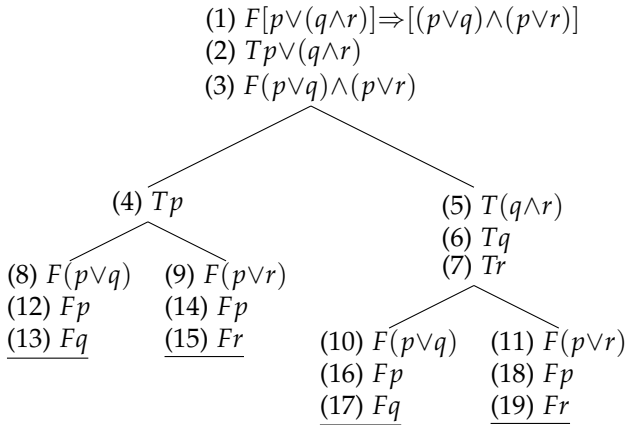
Thus the truth value of  $TX$  is the same as that of  $X$ ; the truth value of  $FX$  is the same as that of  $\sim X$ .

By the *conjugate* of a signed formula we mean the result of changing " $T$ " to " $F$ " or " $F$ " to " $T$ " (thus the conjugate of  $TX$  is  $FX$ ; the conjugate of  $FX$  is  $TX$ ).

### ILLUSTRATION OF THE METHOD OF TABLEAUX

Before we state the eight rules for the construction of tableaux, we shall illustrate the construction with an example.

Suppose we wish to prove the formula  $[p \vee (q \wedge r)] \Rightarrow [(p \vee q) \wedge (p \vee r)]$ . The following tableau does this; the explanation is given immediately following the tableau.



EXPLANATION. The tableau was constructed as follows. We want to derive a contradiction from the assumptions that the formula

$$[p \vee (q \wedge r)] \Rightarrow [(p \vee q) \wedge (p \vee r)]$$

is false. So our first line consists of this formula preceded by the letter  $F$ . Now, a formula of the form  $X \Rightarrow Y$  can be false only if  $X$  is true and  $Y$  is false. Thus (in the language of tableaux)  $TX$  and  $FY$  are *direct* consequences of the (signed) formula  $F(X \Rightarrow Y)$ . So we write the lines (2) and (3) as *direct* consequences of line (1). Now let us look at line (2); it is of the form  $T(X \vee Y)$ , where  $X = p$ ,  $Y = (q \wedge r)$ . We cannot draw any *direct* conclusion about the truth value of  $X$  nor about the truth value of  $Y$ ; all we can infer is that *either*  $TX$  *or*  $TY$ . So the tableau *branches* into two columns; one for each possibility. Thus line (2) *branches* into lines (4) and (5). Line (5), viz.  $T(q \wedge r)$ , immediately yields  $Tq$  and  $Tr$  as direct consequences; we thus have lines (6) and (7). Now look at (3). It is of the form  $F(X \wedge Y)$ . This means that *either*  $FX$  *or*  $FY$ . We also know that either (4) or (5) holds. So for *each* of the possibilities (4), (5) we have the two possibilities  $FX$ ,  $FY$ . There are hence now four possibilities. So each of the branches (4), (5) branches again into the possibilities  $FX$ ,  $FY$ . More specifically, (4) branches to (8), (9), and (5) branches to (10), (11) (which are respectively the same as (8), (9)). Lines (12), (13) are direct consequences of (8); (14), (15) are direct consequences of (9); (16), (17) of (10); and (18), (19) of (11).

We now look at the leftmost branch and we see that (12) is a direct contradiction of (4) (i.e., it is the conjugate of (4)), so we put a bar after (13) to signify that this branch leads to a contradiction. Similarly, (14) contradicts (4), so we "close" the branch leading to (15) by putting a bar after (15). The next branch is closed by virtue of (17) and (6). Finally, the rightmost branch is closed by virtue of (19) and (7). Thus all branches

lead to a contradiction, so line (1) is untenable. Thus

$$[p \vee (q \wedge r)] \Rightarrow [(p \vee q) \wedge (p \vee r)]$$

can never be false in any interpretation, so it is a tautology.

REMARK. The numbers to the left of the lines are only for the purpose of identification in the above explanation; we do not need them for the actual construction.

REMARK. We could have closed some of our branches a bit earlier; lines (13), (15) are superfluous. In subsequent examples we shall close a branch as soon as a contradiction appears (a contradiction is of the form of two formulas  $FX$ ,  $TX$ ).

### RULES FOR THE CONSTRUCTION OF TABLEAUX

We now state all the rules in schematic form; explanations immediately follow. For each logical connective, there are two rules: one for a formula preceded by  $T$ , the other for a formula preceded by  $F$ :

- |                                                                                              |                                                                                      |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>(1) <math>\frac{T \sim X}{FX}</math></p>                                                  | <p><math>\frac{F \sim X}{TX}</math></p>                                              |
| <p>(2) <math>\frac{T(X \wedge Y)}{\begin{array}{c} TX \\ TY \end{array}}</math></p>          | <p><math>\frac{F(X \wedge Y)}{\begin{array}{cc} FX &amp; FY \end{array}}</math></p>  |
| <p>(3) <math>\frac{T(X \vee Y)}{\begin{array}{cc} TX &amp; TY \end{array}}</math></p>        | <p><math>\frac{F(X \vee Y)}{\begin{array}{c} FX \\ FY \end{array}}</math></p>        |
| <p>(4) <math>\frac{T(X \Rightarrow Y)}{\begin{array}{cc} FX &amp; TY \end{array}}</math></p> | <p><math>\frac{F(X \Rightarrow Y)}{\begin{array}{c} TX \\ FY \end{array}}</math></p> |

SOME EXPLANATIONS. Rule (1) means that from  $T \sim X$  we can directly infer  $FX$  (in the sense that we can subjoin  $FX$  to any branch passing through  $T \sim X$ ) and that from  $F \sim X$  we can directly infer  $TX$ . Rule (2) means that  $T(X \wedge Y)$  directly yields both  $TX$  and  $TY$ , whereas  $F(X \wedge Y)$  *branches* into  $FX$  and  $FY$ . Rules (3) and (4) can now be understood analogously.

Signed formulas, other than signed variables, are of two types: (A) those that have *direct* consequences (viz.  $F \sim T$ ,  $T \sim X$ ,  $T(X \wedge Y)$ ,  $F(X \vee Y)$ ,  $F(X \Rightarrow Y)$ ) and (B) those that *branch* (viz.  $F(X \wedge Y)$ ,  $T(X \vee Y)$ ,  $T(X \Rightarrow Y)$ ).

It is practically desirable, in constructing a tableau, that when a line of type (A) appears on the tableau, we simultaneously subjoin its consequences to *all* branches that pass through the line. Then that line

need never be used again. And in using a line of type (B), we divide *all* branches that pass through the line into sub-branches, and the line need never be used again. For example, in Tableau 1 below, we use (1) to get (2) and (3), and (1) is never used again. From (2) we get (4) and (5), and (2) is never used again. Line (3) yields (8), (9), (10), (11), and (3) is never used again, etc.

If we construct a tableau in the above manner, it is not difficult to see that after a finite number of steps we must reach a point where every line has been used (except, of course, for signed variables, which are never used at all to create new lines). At this point our tableau is *complete* (in a precise sense that we will subsequently define).

One way to complete a tableau is to work systematically downward, i.e., never to use a line until all lines above it (on the same branch) have been used. Instead of this procedure, however, it turns out to be more efficient to give priority to lines of type (A)—i.e., to use up all such lines at hand before using those of type (B). In this way, one will avoid repeating the same formula on different branches; rather, it will have only one occurrence *above* all those branch points.

As an example of both procedures, let us prove the formula

$$[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)].$$

Tableau 1 works systematically downward; Tableau 2 uses the second suggestion. For the convenience of the reader, we put to the right of each line the number of the line from which it was inferred.

It is apparent that Tableau 2 is quicker to construct than Tableau 1, involving 13 rather than 23 lines.

As another practical suggestion, one might put a checkmark to the right of a line as soon as it has been used. This will subsequently aid the eye in hunting upward for lines that have not yet been used. (The checkmark may be later erased, if the reader so desires.)

The method of tableaux can also be used to show that a given formula is a logical consequence of a given finite set of formulas. Suppose we wish to show that  $X \Rightarrow Z$  is a logical consequence of the two formulas  $X \Rightarrow Y$ ,  $Y \Rightarrow Z$ . We could, of course, simply show that  $[(X \Rightarrow Y) \wedge (Y \Rightarrow Z)] \Rightarrow (X \Rightarrow Z)$  is a tautology by constructing a truth table for it. Alternatively, we can construct a tableau starting with the three signed formulas

$$\begin{array}{l} T(X \Rightarrow Y), \\ T(Y \Rightarrow Z), \\ F(X \Rightarrow Z), \end{array}$$

and show that all branches close.

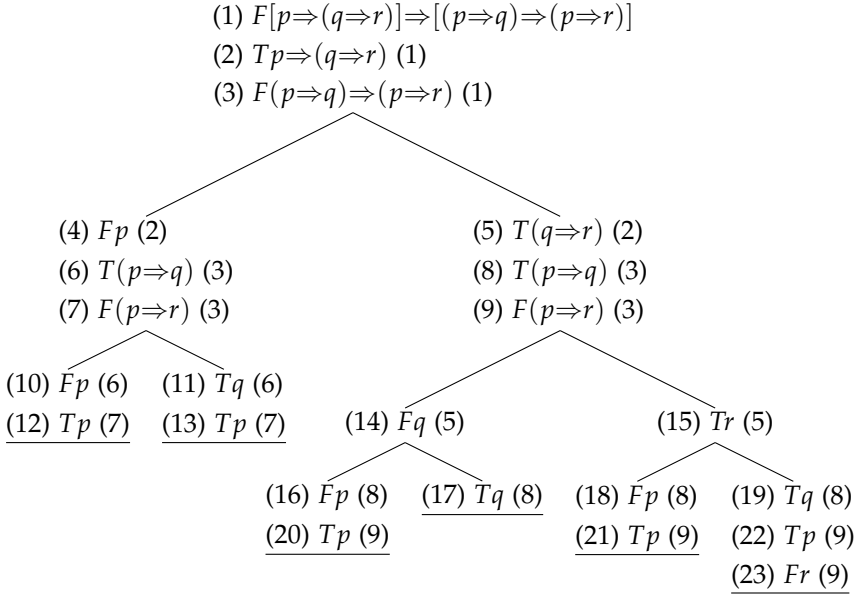


TABLEAU 1.

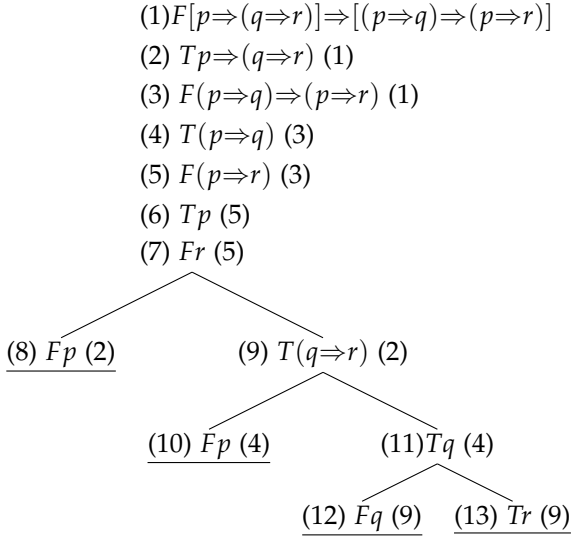


TABLEAU 2.

In general, to show that  $Y$  is logically implied by  $X_1, \dots, X_n$ , we can construct either a closed tableau starting with  $F[(X_1 \wedge \dots \wedge X_n) \Rightarrow Y]$  or one starting with

$$\begin{array}{c} TX_1 \\ \vdots \\ TX_n \\ FY. \end{array}$$

### TABLEAUX USING UNSIGNED FORMULAS

Our use of the letters  $T$  and  $F$ , though perhaps heuristically useful, is theoretically quite dispensable—simply delete every  $T$  and substitute  $\sim$  for  $F$ . (In which case, incidentally, the first half of Rule (1) becomes superfluous.) The rules then become

- $$\begin{array}{ll} (1) & \frac{\sim\sim X}{X} \\ (2) & \frac{X \wedge Y}{\begin{array}{c} X \\ Y \end{array}} \quad \frac{\sim(X \wedge Y)}{\begin{array}{cc} \sim X & \sim Y \end{array}} \\ (3) & \frac{X \vee Y}{\begin{array}{cc} X & Y \end{array}} \quad \frac{\sim(X \vee Y)}{\begin{array}{c} \sim X \\ \sim Y \end{array}} \\ (4) & \frac{X \Rightarrow Y}{\begin{array}{cc} \sim X & Y \end{array}} \quad \frac{\sim(X \Rightarrow Y)}{\begin{array}{c} X \\ \sim Y \end{array}} \end{array}$$

In working with tableaux that use unsigned formulas, “closing” a branch means terminating the branch as soon as two formulas appear, one of which is the *negation* of the other. A tableau is called *closed* if every branch is closed. A branch is called *open* if it is not closed.

By a tableau *for* a formula  $X$ , we mean a tableau that starts with  $X$ . If we wish to prove a formula  $X$  to be a tautology, we construct a tableau not for the formula  $X$  but for its negation  $\sim X$ .

**EXERCISE 11.1.** By the tableau method, prove the following tautologies:

- (a)  $q \Rightarrow (p \Rightarrow q)$ .
- (b)  $((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$ .
- (c)  $((p \Rightarrow r) \wedge (q \Rightarrow r)) \Rightarrow ((p \vee q) \Rightarrow r)$ .

$$(d) ((p \Rightarrow q) \wedge (p \Rightarrow r)) \Rightarrow (p \Rightarrow (q \wedge r)).$$

$$(e) \sim(p \wedge q) \Rightarrow (\sim p \vee \sim q).$$

$$(f) \sim(p \vee q) \Rightarrow (\sim p \wedge \sim q).$$

$$(g) (p \wedge (q \vee r)) \Rightarrow ((p \wedge q) \vee (p \wedge r)).$$

$$(h) (p \vee (q \wedge r)) \Rightarrow ((p \vee q) \wedge (p \vee r)).$$

$$(i) ((p \Rightarrow q) \wedge (\sim p \Rightarrow q)) \Rightarrow q.$$

$$(j) (((p \wedge q) \Rightarrow r) \wedge (p \Rightarrow (q \vee r))) \Rightarrow (p \Rightarrow r).$$

## A UNIFYING NOTATION

In this book, we are taking  $\sim$ ,  $\wedge$ ,  $\vee$  and  $\Rightarrow$  as independent logical connectives—the so-called starters—even though we could have defined some of them in terms of the others. Some treatments of propositional logic take only  $\sim$  and  $\Rightarrow$  as starters and define the others in terms of them; other treatments start with only  $\sim$  and  $\vee$ ; others with  $\sim$  and  $\wedge$ ; others with  $\sim$ ,  $\wedge$  and  $\vee$ ; still other with all four of  $\sim$ ,  $\wedge$ ,  $\vee$ ,  $\Rightarrow$ —which is what we are doing. The advantage of using many starters is that proofs *within* the system tend to be shorter and more natural; but proofs *about* the system (the *metatheory*, which we study later on) tend to be much longer and involve analyzing many different cases. On the other hand, those systems that use few starters have advantages in the quickened metatheory, but proofs *within* the system tend to be long and unnatural. Well, in [17] was presented a unifying scheme that enables one, so to speak, to have one's cake and eat it too—that is, it combines the advantages of many starters with the advantages of few. For example, in our present setup, there are eight types of signed formulas (other than signed variables)—two for each of the four connectives  $\sim$ ,  $\wedge$ ,  $\vee$ ,  $\Rightarrow$ —and hence eight tableau rules. Well, our unified scheme allows us to collapse these eight rules into only two! Moreover, in the next section of this chapter, we will be raising some vital questions about the adequacy of the tableau method (questions about the metatheory) where answers would require an analysis of eight separate cases, were it not for the unifying scheme, which makes it possible to reduce the number of cases to two. Does this whet your appetite? Well, here is the scheme (which will play a vital role throughout this volume).

We use the Greek letter  $\alpha$  to stand for any signed formula of one of the five forms  $TX \wedge Y$ ,  $FX \vee Y$ ,  $FX \Rightarrow Y$ ,  $T \sim X$ ,  $F \sim X$ . For every such formula  $\alpha$ , we define the two signed formulas  $\alpha_1$  and  $\alpha_2$  as follows:

|                                    |      |                                           |
|------------------------------------|------|-------------------------------------------|
| if $\alpha = T(X \wedge Y)$ ,      | then | $\alpha_1 = TX$ and $\alpha_2 = TY$ ;     |
| if $\alpha = F(X \vee Y)$ ,        | then | $\alpha_1 = FX$ and $\alpha_2 = FY$ ;     |
| if $\alpha = F(X \Rightarrow Y)$ , | then | $\alpha_1 = TX$ and $\alpha_2 = FY$ ;     |
| if $\alpha = T \sim X$ ,           | then | $\alpha_1$ and $\alpha_2$ are both $FX$ ; |
| if $\alpha = F \sim X$ ,           | then | $\alpha_1$ and $\alpha_2$ are both $TX$ . |

For perspicuity, we summarize these definitions in the following table:

| $\alpha$             | $\alpha_1$ | $\alpha_2$ |
|----------------------|------------|------------|
| $T(X \wedge Y)$      | $TX$       | $TY$       |
| $F(X \vee Y)$        | $FX$       | $FY$       |
| $F(X \Rightarrow Y)$ | $TX$       | $FY$       |
| $T \sim X$           | $FX$       | $FX$       |
| $F \sim X$           | $TX$       | $TX$       |

We refer to  $\alpha_1$  and  $\alpha_2$  as the *components* of  $\alpha$ . We note that under any interpretation,  $\alpha$  is true if and only if  $\alpha_1$  and  $\alpha_2$  are *both* true. Accordingly, we refer to an  $\alpha$ -formula as one of *conjunctive* type, or of *type A*.

We let  $\beta$  be any signed formula of one of the five forms  $FX \wedge Y$ ,  $TX \vee Y$ ,  $TX \Rightarrow Y$ ,  $T \sim X$ ,  $F \sim X$ , and we define its components  $\beta_1$  and  $\beta_2$  as in the following table:

| $\beta$              | $\beta_1$ | $\beta_2$ |
|----------------------|-----------|-----------|
| $T(X \vee Y)$        | $TX$      | $TY$      |
| $F(X \wedge Y)$      | $FX$      | $FY$      |
| $T(X \Rightarrow Y)$ | $FX$      | $TY$      |
| $T \sim X$           | $FX$      | $FX$      |
| $F \sim X$           | $TX$      | $TX$      |

We note that, under any interpretation,  $\beta$  is true if and only if either  $\beta_1$  is true or  $\beta_2$  is true, and we accordingly refer to a  $\beta$ -formula as one of *disjunctive* type, or of *type B*.

Some special remarks are in order for signed formulas of the form  $T \sim X$  or  $F \sim X$ : These are the only types of formulas that we have classified as of both type  $\alpha$  and type  $\beta$ , because, in both cases, its two components are identical, and therefore to say that both components are true is equivalent to saying that at least one of them is true—so we can regard them either as an  $\alpha$  or as a  $\beta$ .<sup>1</sup>

Using our unifying  $\alpha$ ,  $\beta$ -notation, we note the pleasant fact that our eight tableau rules can now be succinctly lumped into the following two:

|         |                             |         |                         |
|---------|-----------------------------|---------|-------------------------|
| RULE A. | $\alpha$                    | RULE B. | $\beta$                 |
|         | $\frac{\alpha_1}{\alpha_2}$ |         | $\beta_1 \quad \beta_2$ |

<sup>1</sup>In [17], we took them only as  $\alpha$ , but the present scheme works better for this volume.

REMARK. Of course, in constructing a tableau, we should treat  $T\sim X$  and  $F\sim X$  as  $\alpha$ 's, since it is pointless to extend a branch to two identical branches! More generally, given any  $\beta$ -type formula whose two components are identical (such as  $TX\vee X$ ), one should simply adjoin a single copy of the component rather than branch to two identical copies—in our case, adjoin  $TX$  rather than branch to  $TX$  and  $TX$ !

### SOME PROPERTIES OF CONJUGATION

We recall that by the conjugate of  $TX$  we mean  $FX$ , and by the conjugate of  $FX$  we mean  $TX$ . We shall use the notation  $\overline{X}$  to mean the *conjugate* of  $X$  (thus  $\overline{TX} = FX$ , and  $\overline{FX} = TX$ ). Conjugation obeys the following nice symmetric laws:

- $J_0$ . (a)  $\overline{\overline{X}}$  is distinct from  $X$ ;  
 (b)  $\overline{\overline{\overline{X}}} = X$ .

- $J_1$ . The conjugate of any  $\alpha$  is some  $\beta$ , and the conjugate of any  $\beta$  is an  $\alpha$ .

- $J_2$ . If  $\beta$  is the conjugate of  $\alpha$ , then  $\beta_1$  is the conjugate of  $\alpha_1$  and  $\beta_2$  is the conjugate of  $\alpha_2$ .

Laws  $J_0$  and  $J_1$  are obvious. As to  $J_2$ , this must be verified case by case. For example, consider the case where  $\alpha = (FX \Rightarrow Y)$  and  $\beta = (TX \Rightarrow Y)$ . Then  $\alpha_1 = TX$  and  $\beta_1 = FX$ , which is the conjugate of  $\alpha_1$ . Also  $\alpha_2 = FY$  and  $\beta_2 = TY$ , and so  $\beta_2$  is the conjugate of  $\alpha_2$ .

EXERCISE 11.2. Verify law  $J_2$  for the remaining cases.

- (a)  $\alpha = (TX \wedge Y)$ ,  $\beta = (FX \wedge Y)$ .  
 (b)  $\alpha = (TX \vee Y)$ ,  $\beta = (TX \vee Y)$ .

We can also use a unifying  $\alpha$ - $\beta$  notation for *unsigned* formulas—just delete  $T$  and replace  $F$  by  $\sim$ . The tables then become:

| $\alpha$                | $\alpha_1$ | $\alpha_2$ | $\beta$            | $\beta_1$ | $\beta_2$ |
|-------------------------|------------|------------|--------------------|-----------|-----------|
| $X \wedge Y$            | $X$        | $Y$        | $\sim(X \wedge Y)$ | $\sim X$  | $\sim Y$  |
| $\sim(X \vee Y)$        | $\sim X$   | $\sim Y$   | $X \vee Y$         | $X$       | $Y$       |
| $\sim(X \Rightarrow Y)$ | $X$        | $\sim Y$   | $X \Rightarrow Y$  | $\sim X$  | $Y$       |
| $\sim\sim X$            | $X$        | $X$        | $\sim\sim X$       | $X$       | $X$       |

Now, what about *conjugation* for unsigned formulas? How should we define it? Well, we will take  $X \wedge Y$  and  $\sim(X \wedge Y)$  to be conjugates of each other, and similarly with  $X \vee Y$  and  $\sim(X \vee Y)$ , and similarly with  $X \Rightarrow Y$

and  $\sim(X \Rightarrow Y)$ ; but now the delicate case is that of formulas of the form  $\sim\sim X$ , or  $\sim\sim\sim X$ , or any other formula involving multiple negations. We want our definition of conjugation to be such that law  $J_0$  (b) holds ( $\overline{\overline{X}} = X$ ). Well, the following scheme works: Let  $Y$  be any formula that is not itself a negation of any formula, and let  $X$  be either  $Y$  or  $Y$  preceded by one or more negations (i.e.,  $X=Y$ , or  $\sim Y$ , or  $\sim\sim Y$ , or  $\sim\sim\sim Y$ , etc.). If the number of negations is even, then take  $\overline{X}$  to be the result of adding one more negation, but if the number is odd, then take away one negation. So, for example, if  $Y$  is not itself a negation, then  $\overline{\overline{Y}}=Y$ ,  $\overline{\sim Y}=Y$ ,  $\overline{\sim\sim Y}=\sim\sim Y$ ,  $\overline{\sim\sim\sim Y}=\sim\sim Y$ , etc. In this way,  $\overline{\overline{X}}$  is always  $X$ . Indeed, laws  $J_0$ ,  $J_1$  and  $J_2$  all hold under this definition of conjugation for unsigned formulas.

DISCUSSION. There are other logical connectives that we could have taken as starters that would fit nicely into our unified  $\alpha$ - $\beta$  scheme—for example,  $\downarrow$  (joint denial) and  $|$  (the Sheffer stroke). (Recall that  $p \downarrow q$  means that  $p$  and  $q$  are both false, and  $p | q$  means that  $p$  and  $q$  are not both true.)

PROBLEM 11.1. Suppose we had taken these two as independent connectives. What should be the tableau rules for  $TX \downarrow Y$ ,  $FX \downarrow Y$ ,  $TX | Y$ ,  $FX | Y$ ? Also, which of them are  $\alpha$ 's and which are  $\beta$ 's?

### THE BICONDITIONAL

We have not given tableau rules for the biconditional  $\equiv$  (if and only if), because we could simply take  $X \equiv Y$  as an abbreviation of

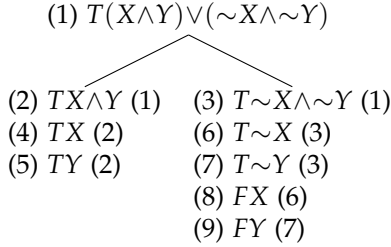
$$(X \wedge Y) \vee (\sim X \wedge \sim Y)$$

(or alternatively of the equivalent formula  $(X \Rightarrow Y) \wedge (Y \Rightarrow X)$ ). If we alternatively take  $\equiv$  as an independent connective, then the obviously appropriate tableau rules are the following:

|                                                                                                                           |                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| RULE A. $TX \equiv Y$<br>$\begin{array}{cc} \diagup & \diagdown \\ TX & FX \\ \diagdown & \diagup \\ TY & FY \end{array}$ | RULE B. $FX \equiv Y$<br>$\begin{array}{cc} \diagup & \diagdown \\ TX & FX \\ \diagdown & \diagup \\ FY & TY \end{array}$ |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

(To say that  $X \equiv Y$  is true is to say that  $X$  and  $Y$  are both true or both false. To say that  $X \equiv Y$  is false is to say that one of them is true and the other false, thus either  $TX$  and  $FY$  or  $FX$  and  $TY$ .)

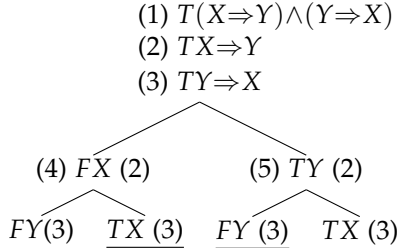
Let us note that if we took  $X \equiv Y$  as an abbreviation of  $(X \wedge Y) \vee (\sim X \wedge \sim Y)$ , our tableau for the T-case would have gone as follows:



It really leads to the same thing, since on the left branch we have  $TX$  and  $TY$ , and on the right branch  $FX$  and  $FY$ .

If a tableau is constructed for  $F(X \wedge Y) \vee (\sim X \wedge \sim Y)$ , it is easily seen that one branch will contain  $TX$  and  $FY$  and the other  $FX$  and  $TY$ .

Suppose we instead took  $X \equiv Y$  as an abbreviation of  $(X \Rightarrow Y) \wedge (Y \Rightarrow X)$ . Let us see what the tableau for  $T(X \Rightarrow Y) \wedge (Y \Rightarrow X)$  looks like.



We now have four branches, but only two *open* ones, and again one of them contains  $FX$  and  $FY$  and the other,  $TX$  and  $TY$ .

(The reader should now try to make a tableau for  $F(X \Rightarrow Y) \wedge (Y \Rightarrow X)$ .)

We thus see that it makes little difference whether we take  $\equiv$  independent of the other connectives or not, but if we do, and use the rule given above, it eliminates unnecessary work.

We remark that very few (if any) treatments of propositional logic take  $\equiv$  as a starter, and we also remark that  $\equiv$  does not fit at all into our unifying  $\alpha$ - $\beta$  scheme.

**EXERCISE 11.3.** By tableau, prove the following:

- $p \equiv (q \equiv r) \equiv ((p \equiv q) \equiv r)$ .
- $((p \Rightarrow q) \wedge (q \Rightarrow p)) \equiv ((p \wedge q) \vee (\sim p \wedge \sim q))$ .
- $(p \equiv (p \Rightarrow q)) \equiv (p \wedge q)$ .
- $(q \equiv (p \Rightarrow q)) \equiv (p \vee q)$ .
- $(p \Rightarrow q) \equiv (p \equiv (p \wedge q))$ .

## SOME VITAL QUESTIONS: PROLEGOMENA TO METATHEORY

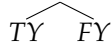
A proof procedure for propositional logic is called *correct* if it proves only tautologies and *complete* if it can prove all tautologies. The truth-table method is obviously both correct and complete. As to the tableau method, it is easy to show that it is correct, but the key question is whether the method is complete. That is, if  $X$  is a tautology, can we be sure that we can get a tableau starting with  $FX$  to close? Before the reader answers this question too hastily, we wish to point out that if we delete some of the rules for the construction of tableaux, it will still be true that a *closed* tableau for  $FX$  *always* indicates that  $X$  is a tautology, but if we delete too many of the rules, then we may not have enough power to derive a closed tableau for  $FX$  whenever  $X$  is a tautology. As examples, if we delete the first half of the conjunction rule (that from  $TX \wedge Y$  we can directly infer both  $TX$  and  $TY$ ) then it would be impossible to prove the tautology  $(p \wedge q) \Rightarrow p$ , though it would still be possible to prove  $p \Rightarrow (q \Rightarrow (p \wedge q))$ . On the other hand, if we delete the second half of the rule but retain the first half, then we could prove  $(p \wedge q) \Rightarrow p$  but we couldn't prove  $p \Rightarrow (q \Rightarrow (p \wedge q))$ . So the question is whether we really have enough rules to prove *all* tautologies! The answer to that question is not so obvious, and will be given in a later chapter. (Meanwhile the more ambitious reader might like to try his or her hand at it: Can you either prove that the tableau method *is* complete, or find a tautology that is *not* provable by the method?)

### A RELATED QUESTION

Given two formulas  $X$  and  $Y$ , suppose that  $X$  and  $X \Rightarrow Y$  are both provable by the tableau method. Does it necessarily follow that  $Y$  is provable by this method? Of course, if  $X$  and  $X \Rightarrow Y$  are both tautologies, so is  $Y$  (because  $Y$  is true under all interpretations in which  $X$  and  $X \Rightarrow Y$  are both true, so if  $X$  and  $X \Rightarrow Y$  are both true under all interpretations, so is  $Y$ ). Therefore, since the tableau method is correct, if  $X$  and  $X \Rightarrow Y$  are both provable by tableau, then both are tautologies, hence  $Y$  is also a tautology—but this is as far as we can go, since we have not yet answered the question of whether all tautologies are provable by tableau! If it turns out that the method is complete, it would indeed then follow that  $Y$  is provable (if  $X$  and  $X \Rightarrow Y$  both are), but what if the method is *not* complete?

## SYNTHETIC TABLEAUX

Suppose we add to our tableau rules the following:



This means that at any stage of the construction of a tableau, we can pull an arbitrary formula out of the air, and then let any (or all) of the branches split to  $TY$  and  $FY$  (or we can even start a tableau with the two branches  $TY$  and  $FY$ ).

Let us call this rule the *splitting rule*, and let us call a tableau that uses this rule a *synthetic tableau* (and we will call *analytic* those tableaux that do not use the splitting rule). This rule is certainly sound, since, in any interpretation, one (and only one) of the signed formulas  $TY$  and  $FY$  is true, and so any formula provable by a synthetic tableau is also a tautology. Now an interesting question arises: Can one prove any more formulas with the splitting rule than without it? Does the addition of the splitting rule increase the class of provable formulas? Is it possible that without the splitting rule, the tableau method is incomplete, but adding this rule makes the system complete? All these questions will be answered later on.

One thing I will tell you now: We previously raised the question of whether if  $X$  and  $X \Rightarrow Y$  are both provable by the tableau method,  $Y$  is necessarily provable, and the answer was not immediate. For synthetic tableaux, however, the answer *is* immediate and affirmative.

**PROBLEM 11.2.** Why is it true that for synthetic tableaux, if  $X$  and  $X \Rightarrow Y$  are both provable, so is  $Y$ ? More constructively, given a closed tableau  $\mathfrak{T}_1$  for  $FX$  and a closed tableau  $\mathfrak{T}_2$  for  $FX \Rightarrow Y$ , how would you combine them (using the splitting rule) to obtain a closed tableau for  $FY$ ?

## ANOTHER RELATED QUESTION

In mathematics, it is customary in proving a theorem  $X$  to use in the proof a previously proved theorem  $Y$ . Suppose we incorporate this into our tableau system by adding the rule that at any stage of the construction of a tableau, one may, on any branch, introduce  $TY$ , where  $Y$  is any formula previously proved. Does this increase the class of provable formulas? Well, without the splitting rule the answer is not obvious, and will be given in a later chapter. But with the splitting rule, the answer is easily seen to be *no*.

**PROBLEM 11.3.** Why is this?

## SOLUTIONS

11.1. The tableau rules for  $\downarrow$  and  $|$  are the following:

$$\frac{TX\downarrow Y}{\frac{FX}{FY}} \quad \frac{FX|Y}{\frac{TX}{TY}} \quad \frac{T(X|Y)}{FX \quad FY} \quad \frac{F(X\downarrow Y)}{TX \quad TY}$$

Thus  $TX\downarrow Y$  and  $FX|Y$  are the  $\alpha$ 's, and the others are the  $\beta$ 's.

11.2. We are given a closed tableau  $\mathfrak{T}_1$  for  $FX$  and a closed tableau  $\mathfrak{T}_2$  for  $FX\Rightarrow Y$ , and we construct a closed synthetic tableau for  $FY$  as follows:

$$\begin{array}{c} FY \\ \swarrow \quad \searrow \\ TX\Rightarrow Y \quad \underline{\underline{FX\Rightarrow Y}} \\ \swarrow \quad \searrow \\ \underline{\underline{FX}} \quad \underline{\underline{TY}} \end{array}$$

EXPLANATION. We start the tableau with  $FY$ . We then immediately split to  $TX\Rightarrow Y$  (left branch) and  $FX\Rightarrow Y$  (right branch). The double bar under  $FX\Rightarrow Y$  indicates that one *can* get a closure under it (by using tableau  $\mathfrak{T}_2$ ). Next, the left branch  $TX\Rightarrow Y$  itself branches to  $FX$  and  $TY$ . The  $TY$  clashes with  $FY$  and that sub-branch closes. The  $FX$  extends to the tableau  $\mathfrak{T}_1$ , which is closed.

11.3. We are given a proof of a formula  $Y$ , a closed tableau  $\mathfrak{T}$  whose origin is  $FY$ . We wish to prove a formula  $X$ . At any stage of a tableau starting with  $FX$ , in place of arbitrarily adding  $TY$  as a branch, if we use the splitting rule, then we can alternatively split the branch thus:

$$\begin{array}{c} \swarrow \quad \searrow \\ TY \quad \underline{\underline{FY}} \end{array}$$

But we can then close  $FY$  by copying the tableau  $\mathfrak{T}$ . So, in effect, we have added  $TY$ .



## - CHAPTER 12 -

# ALL AND SOME

Propositional logic, which we have just studied, is only the *beginning* of the logic we need for mathematics and science. The real meat comes in the field known as *first-order logic*, which is one of the main subjects of this book. It uses the logical connectives of propositional logic, with the addition of the notions of *all* and *some* (“some” in the sense of *at least one*), which will be symbolically treated in the next chapter. In this chapter, we treat these two notions on an informal and intuitive level.

To begin with, suppose a member of a certain club says: “All Frenchmen in this club wear hats.” Now, suppose it turns out that there are no Frenchmen in the club, then how would you regard the statement—*true*, *false*, or *neither*? I guess many would say *false*, many would say *neither* (in other words, *inapplicable*, or *meaningless*) and perhaps some would say *true*. Well, it will shock many of you to hear that in logic, mathematics and computer science, the statement is regarded as *true*! The statement means nothing more nor less than that there are no Frenchmen in the club who don’t wear hats. Thus if there are no Frenchmen at all in the club, then there are certainly no Frenchmen in the club who don’t wear hats, and so the statement is true.

In general, the statement “All  $A$ ’s are  $B$ ’s” is regarded as automatically true (“*vacuously true*” is the technical term) if there are no  $A$ ’s. So, for example, shocking as it may seem, the following statement is true:

All unicorns have five legs.

The only way the above sentence can be falsified is to exhibit at least one unicorn who doesn’t have five legs, which is not possible, since there

are no unicorns. It is also true that all unicorns have six legs! *Anything* one says about *all* unicorns is vacuously true. On the other hand, the statement that all *horses* have six legs is easily shown to be false; just exhibit a horse with only four legs!

I realize that all this takes some getting used to, but, in the next chapter, we will see the practical use of this way of looking at it. For now, we will consider some knight-knave problems involving the notions of *all* and *some*.

## PROBLEMS

Abercrombie once visited a whole cluster of knight-knave islands of the simple type of Chapter 1, in which all knights tell the truth and all knaves lie. He was interested in the proportions of knights to knaves on the islands and, also, whether there was any correlation between lying and smoking.

PROBLEM 12.1. On the first island he visited, all the inhabitants said the same thing: "All of us here are of the same type."

What can be deduced about the inhabitants of that island?

PROBLEM 12.2. On the next island, all the inhabitants said: "Some of us are knights and some are knaves."

What is the composition of that island?

PROBLEM 12.3. On the next island, Abercrombie one day interviewed all the inhabitants but one, who was asleep. They all said: "All of us are knaves." The next day, Abercrombie met the inhabitant who was asleep the day before, and asked him: "Is it true that all the inhabitants of this island are knaves?" The inhabitant then answered (*yes* or *no*). What answer did he give?

PROBLEM 12.4. On the next island, Abercrombie was particularly interested in the smoking habits of the natives. They all said the same thing: "All knights on this island smoke."

What can be deduced about the knight-knave distributions and the smoking habits of the natives?

PROBLEM 12.5. On the next island, each one said: "Some knaves on this island smoke."

What can be deduced from this?

PROBLEM 12.6. On the next island all were of the same type, and each one said: "If I smoke, then all the inhabitants of this island smoke."

What can be deduced?

PROBLEM 12.7. On the next island again all were of the same type, and each one said: "If any inhabitant of this island smokes, then I smoke."

What can be deduced?

PROBLEM 12.8. On the next island, too, all were of the same type, and each one said: "Some of us smoke, but I don't."

What follows?

PROBLEM 12.9. Suppose that on the same island, instead, each inhabitant made the following two statements: "Some of us smoke." "I don't smoke."

What would you conclude?

PROBLEM 12.10. The next island visited by Abercrombie was inhabited by two tribes—Tribe A and Tribe B. All the members of Tribe A said: "All the inhabitants of this island are knights." "All of us smoke."

Each member of Tribe B said: "Some of the inhabitants of this island are knaves." "No one on this island smokes."

What can be deduced?

PROBLEM 12.11. On this island, as well as the next two visited by Abercrombie, there are male and female knights and knaves. The female knights lie and the female knaves tell the truth, whereas the males act as before (male knights are truthful and male knaves are not). All the inhabitants of this island (male and female) said the same thing: "All inhabitants of this island are knights."

What can be deduced?

PROBLEM 12.12. On the next island, all the men said: "All the inhabitants of this island are knaves." Then the women were asked whether it was true that all the inhabitants were knaves. They all gave the same answer (*yes* or *no*).

What answer did they give?

PROBLEM 12.13. On the next island, all the men said: "All the inhabitants are knights and they all smoke." The women all said: "Some of the inhabitants are knaves. All the inhabitants smoke."

What can be deduced?

PROBLEM 12.14. On the next island visited by Abercrombie, he met six natives, named Arthur, Bernard, Charles, David, Edward, and Frank, who made the following statements:

- Arthur: Everyone here smokes cigarettes.  
Bernard: Everyone here smokes cigars.  
Charles: Everyone here smokes either cigarettes  
or cigars or both.  
David: Arthur and Bernard are not both knaves.  
Edward: If Charles is a knight, so is David.  
Frank: If David is a knight, so is Charles.

Is it possible to determine of any one of these that he is a knight, and if so, which one or ones?

PROBLEM 12.15. It has been related that one day a god came down from the sky and classified each inhabitant of the earth as either *special* or *non-special*. As it turns out, for each person  $x$ ,  $x$  was special if and only if it was the case that either everyone was special or no one was special. Which of the following three statements logically follows from this?

- (1) No one is special.
- (2) Some are special and some are not.
- (3) Everyone is special.

PROBLEM 12.16. According to *another* version of the above story, it turned out that for each person  $x$ ,  $x$  was special if and only if some of the people were special and some were not. If this version is correct, then which of the above statements (1), (2), (3) logically follow?

PROBLEM 12.17. On a certain planet, each inhabitant was classified as either *good* or *evil*. A statistician from our planet visited that planet and came to the correct conclusion that for each inhabitant  $x$ ,  $x$  was good if and only if it was the case that all the good inhabitants had green hair. Which of the following three statements logically follow?

- (1) All of them are good.
- (2) None of them are good.
- (3) Some of them are good and some are not.

Also, which of the following three statements follow?

- (4) All of them have green hair.
- (5) None of them have green hair.
- (6) Some of them have green hair and some do not.

PROBLEM 12.18. On another planet, again each inhabitant is classified as either *good* or *evil*. It turns out that for each inhabitant  $x$ ,  $x$  is good if and only if there is at least one evil inhabitant who has green hair. Which of (1)–(6) above logically follows?

PROBLEM 12.19. On a certain island there is a barber named Jim who shaves all those inhabitants who don't shave themselves. Does Jim shave himself or doesn't he?

PROBLEM 12.20. On another island there is a barber named Bill who shaves *only* those who don't shave themselves. (In other words, he never shaves an inhabitant who shaves himself.) Does Bill shave himself or doesn't he?

PROBLEM 12.21. What would you say about an island on which there is a barber who shaves *all those* and *only those* inhabitants who don't shave themselves? (In other words, if an inhabitant shaves himself, the barber won't shave him, but if the inhabitant doesn't shave himself, then the barber shaves him.) Does this barber shave himself or doesn't he? What would you say about such a barber?

NOTE. Be sure to read the important discussion following the solution!

PROBLEM 12.22 (VALID AND SOUND SYLLOGISMS). A syllogism is an argument consisting of a major premise, a minor premise and a conclusion. A syllogism is called *valid* if the conclusion really does follow from the premises, regardless of whether the premises themselves are true. A syllogism is called *sound* if it is valid and if the premises themselves are true. For example, the following syllogism is sound:

|                       |                 |
|-----------------------|-----------------|
| All men are mortal.   | (major premise) |
| Socrates is a man.    | (minor premise) |
| ∴ Socrates is mortal. | (conclusion)    |

(The symbol “∴” abbreviates “therefore.”)

The following syllogism, though obviously not sound, *is* valid!

|                            |
|----------------------------|
| All men have green hair.   |
| Socrates is a man.         |
| ∴ Socrates has green hair. |

This syllogism is unsound, because the major premise (“all men have green hair”) is simply not true. But the syllogism is valid, because if it were really the case that all men had green hair, then Socrates, being a man, would have to have green hair. Both the above syllogisms are special cases of the following general syllogism:

$$\begin{array}{l} \text{All } A\text{'s are } B\text{'s.} \\ x \text{ is an } A. \\ \hline \therefore x \text{ is a } B. \end{array}$$

Is the following syllogism valid?

$$\begin{array}{l} \text{Everyone loves my baby.} \\ \text{My baby loves only me.} \\ \hline \therefore \text{I am my own baby.} \end{array}$$

In the delightfully humorous book *The Devil's Dictionary* by Ambrose Bierce, he gives this example of a syllogism in the following definition of *logic*:

**Logic**, n. The art of thinking and reasoning in strict accordance with the limitations and incapacities of the human misunderstanding. The basis of logic is the syllogism, consisting of a major and minor premise and a conclusion—thus:

*Major Premise:* Sixty men can do a piece of work sixty times as quickly as one man.

*Minor Premise:* One man can dig a posthole in sixty seconds; therefore

*Conclusion:* Sixty men can dig a posthole in one second.

## SOLUTIONS

- 12.1. Since they all said the same thing, they really are all of the same type, so what they said was true. Thus they are all knights.
- 12.2. Since they all said the same thing, then it is not possible that some are knights and some are knaves, hence they all lied. Thus they are all knaves.
- 12.3. All the inhabitants interviewed on the first day said the same thing, so they are all of the same type. They are obviously not knights (no knight would say that all the inhabitants (which includes himself) are knaves), and so they are all knaves. Therefore their statements were all false, and so the sleeping native cannot also be a knave. Since he is a knight, he obviously answered *no*.
- 12.4. Again, all the inhabitants are of the same type (since they all said the same thing). Suppose they are all knaves. Then their statements are false: It is false that all knights on the island smoke. But the only way it can be false is if there is at least one knight on the island who

doesn't smoke, but that is not possible by our assumption that all of them are knaves! So our assumption that they are all knaves leads to a contradiction, and hence they are all knights. It then further follows that all the inhabitants are knights and all of them smoke.

- 12.5. Again, all the natives must be of the same type. If they were knights, they certainly wouldn't say that some knaves on the island smoke (since this would imply that some of them are knaves). Thus they are all knaves, and since their statements are therefore false, it follows that none of them smoke.
- 12.6. We are given that they are all of the same type. Now, consider any one of the natives. He says that if he smokes, then all of them smoke. The only way that it could be false is if he smokes but not all of them smoke. But since they *all* said that, the only way the statements could be false is if each one of the inhabitants smokes, yet not all of them smoke, which is obviously absurd. Therefore the statements cannot be false, and so the inhabitants are all knights. Since their statements are all true, there are two possibilities: (1) None of them smoke (in which case their statements are all true, since a false proposition implies any proposition); (2) All of them smoke. And so all the inhabitants are knights, and all that can be deduced about their smoking is that either none of them smoke or all of them smoke, but there is no way to tell which.
- 12.7. Again, all the inhabitants are of the same type. Each inhabitant claims that if any inhabitant smokes, then he does; and if that were false, then some inhabitant smokes, but the given inhabitant doesn't. Since they *all* say that, it follows that if the statements were false, we would have the contradiction that some inhabitant smokes, but each one doesn't. Thus the natives are all knights, and again either all of them smoke or none of them smoke, and again there is no way to tell which.
- 12.8. Again, they are all of the same type. They couldn't be knights, because if their statements were true, then some of them smoke, yet each one doesn't, which is absurd. Hence they are all knaves. Since their statements are false, it follows that for each inhabitant  $x$ , either it is false that some inhabitants smoke, or it is false that  $x$  doesn't smoke—in other words, either none of them smoke or  $x$  smokes. It could be that none of them smoke. If that alternative doesn't hold, then for each inhabitant  $x$ ,  $x$  smokes, which means that all of them smoke. So the solution is that all of them are knaves, and either all of them smoke or none of them smoke, and there is no way to tell which.

- 12.9. What you *should* conclude is that the situation is impossible!

REASON. If the situation occurred, then for all the same reasons as in the last problem, all the inhabitants would have to be knaves. But this time, for each inhabitant  $x$ , *both* of his statements are false, which means that nobody smokes, yet  $x$  smokes, which is absurd!

(This is another interesting case where a knave can assert the conjunction of two statements, but cannot assert them separately.)

- 12.10. All members of Tribe A are of the same type, and all members of Tribe B are of the same type. Since the members of Tribe B have contradicted the members of Tribe A, it cannot be that the members of both tribes are knights, so the members of Tribe A made false statements, and are therefore knaves. It then further follows that the members of Tribe B are knights, because they correctly said that some of the inhabitants of the island are knaves (and indeed, the members of Tribe A *are* knaves). Then their second statements were also true, hence no one on this island smokes. Thus Tribe A consists of knaves, Tribe B consists of knights, and no one on the island smokes.
- 12.11. Since all the inhabitants said that all the inhabitants are knights, it is impossible that they all are, because if they were, the female knights wouldn't have made the true statement that they are. Therefore all inhabitants of this island lie, and thus the males are all knaves and the females are all knights.
- 12.12. Obviously all the men are knaves (male knights would never say that all the inhabitants are knaves), and their statements are therefore false, which implies that at least one of the women is a knight. And since all the women said the same thing, it follows that all the women are knights. Hence they all falsely answered *yes*.
- 12.13. If the men were knights, then all the inhabitants would be knights and all of them would smoke. But then the female knights would never have made the true statement that they all smoke. Hence the men are all knaves. It further follows that the women are all knaves (because they truthfully said that some of the inhabitants are knaves). Since the women are truthful, all the inhabitants do smoke (as the women said). So all the inhabitants of the island are knaves and they all smoke.
- 12.14. If either everyone there smokes cigarettes or everyone there smokes cigars, then certainly everyone there smokes either cigarettes or cigars. Hence if either Arthur or Bernard is a knight, then so is

Charles. Thus if Arthur and Bernard are not both knaves, then Charles is a knight. Hence if David is a knight, so is Charles (because if David is a knight, then Arthur and Bernard are not both knaves). Thus Frank's statement is true, so Frank is definitely a knight.

Edward's type cannot be determined: It could be that either everyone smokes cigarettes or everyone smokes cigars, in which case it would follow that Edward is a knight (why?), or it could be that everyone smokes either cigarettes or cigars, yet some don't smoke cigarettes and some don't smoke cigars, in which case it would follow that Edward is a knave (why?). Frank is the only one whose type can be determined.

- 12.15. Let  $p$  be the proposition that either everyone is special or no one is special. Also, for each person  $x$ , let us abbreviate the statement that  $x$  is special by  $Sx$ . (In general, in symbolic logic, given any property  $P$  and any individual object  $x$ , the proposition that  $x$  has the property  $P$  is abbreviated  $Px$ .) Recall that two propositions are called *equivalent* if they are either both true or both false. Well, we are given that for each person  $x$ , the proposition  $Sx$  is equivalent to  $p$  ( $x$  is special if and only if  $p$  is true—i.e., if and only if either all or none are special). Then for any two people  $x$  and  $y$ , the propositions  $Sx$  and  $Sy$  must be equivalent to each other, since both of them are equivalent to  $p$ . This means that for any two people, either they are both special or neither one is special, from which it follows that either *all* the people are special, or none of them are special—in other words, the proposition  $p$  is true! Then, since for each person  $x$ , the proposition  $Sx$  is equivalent to  $p$ , it follows that for each person  $x$ , the proposition  $Sx$  is true—in other words, everyone is special!
- 12.16. Again, for each person  $x$ , let  $Sx$  be the proposition that  $x$  is special. Now let  $q$  be the proposition that some people are special and some are not (which is the *opposite* of the proposition that either all or none are special). In the present version,  $Sx$  is equivalent to  $q$ , for each person  $x$ , and so, as in the solution to the last problem, either all are special or none are special (because for every two persons  $x$  and  $y$ , the propositions  $Sx$  and  $Sy$  are equivalent, each being equivalent to  $q$ ). Thus  $q$  is false, and since for any person  $x$ , the proposition  $Sx$  is equivalent to  $q$ ,  $Sx$  must be false. So, according to this version, no one is special.
- 12.17. For each inhabitant  $x$ , let  $Gx$  be the proposition that  $x$  is good. Let  $p$  be the proposition that all the good inhabitants have green hair.

For each inhabitant  $x$ , we are given that  $Gx$  is equivalent to  $p$ , so by the same reasoning as in the last two problems, either all the inhabitants are good, or none of them are good. Suppose that none of them are good. Then for each inhabitant  $x$ , the proposition  $Gx$  is false, and since  $Gx$  is equivalent to  $p$ , it follows that  $p$  must be false. But the only way that  $p$  can be false—the only way that it can be false that all the good inhabitants have green hair—is that there is at least one good inhabitant who doesn't have green hair, which of course implies that at least one inhabitant is good, which is impossible under our assumption that none of the inhabitants are good. Thus the supposition that none of the inhabitants are good leads to a contradiction, and hence must be false. Therefore it is not the case that none of the inhabitants are good, yet we have seen that either all or none are good, hence it must be that all are good. Furthermore, since  $Gx$  is true for each inhabitant  $x$ , and  $Gx$  is equivalent to  $p$ ,  $p$  must be true, which means that every good inhabitant has green hair, and since each inhabitant is good, the conclusion is that all the inhabitants are good and all have green hair.

- 12.18. Again, let  $Gx$  be the proposition that  $x$  is good. Let  $q$  be the proposition that there is at least one evil (not good) inhabitant who has green hair. We are now given that  $Gx$  is equivalent to  $q$ , for each inhabitant  $x$ . Again, it follows that either all the inhabitants are good or none of them are. Suppose that all of them are good. Then for any inhabitant  $x$ , the proposition  $Gx$  holds, and since  $Gx$  is equivalent to  $q$ ,  $q$  must be true—it must be true that at least one evil inhabitant has green hair—and hence there must be at least one evil inhabitant, contrary to the supposition that all the inhabitants are good. Hence the supposition must be false, so none of the inhabitants are good. Furthermore, since  $Gx$  is false for each inhabitant  $x$ , and  $Gx$  is equivalent to  $q$ ,  $q$  must be false. Thus it is not true that at least one evil inhabitant has green hair, yet all the inhabitants are evil, hence none of them have green hair. The conclusion is that all the inhabitants are evil and none of them have green hair.

REMARK. As will be shown in the next chapter, the last four problems above are really Problems 10.1, 10.2, 10.4, and 10.5 over again, but in a new dress.

- 12.19. If Jim didn't shave himself, he would be one of those who don't shave themselves, but Jim must shave all such inhabitants, hence he would have to shave himself, which is a contradiction. Therefore he must shave himself. (Stated otherwise, if he didn't shave himself,

he would fail to shave someone who doesn't shave himself, which goes against the given condition.)

- 12.20. We are given that Bill never shaves anyone who shaves himself. Therefore if Bill shaved himself, he *would* be shaving someone who shaved himself, which is contrary to what is given. Therefore Bill cannot shave himself.
- 12.21. What you *should* say is that the situation is impossible; there *cannot* be such a barber!

REASON. If there were such a barber, then since he shaves all those who don't shave themselves, he must shave himself (Problem 12.19), but on the other hand, since he shaves *only* those who don't shave themselves, he cannot shave himself (Problem 12.20). Thus there can be a barber who shaves *all* those who don't shave themselves, and there can be *another* barber who shaves *only* those who don't shave themselves, but there cannot be a barber who does both.

DISCUSSION. This "Barber Paradox" is a popularization by Bertrand Russell of a famous paradox, also by Russell, that had grave significance for the entire foundation of mathematics. Toward the end of the 19th Century, there appeared a monumental work by Gottlob Frege on set theory that attempted to derive all mathematics from a few basic principles of logic. The main axiom of Frege's system was that, given any property, there existed the set of all things having that property. This seems intuitively evident enough, but Russell showed that it conceals a logical contradiction! Here is Russell's paradox:

Call a set *extraordinary* if it is a member of itself, and *ordinary* if it is not a member of itself. For example, a set of chairs is not itself a chair, thus it is not a member of itself, so is *ordinary*. On the other hand, the set of all entities—if there were such a set—would itself be an entity, hence would be one of its own members, and so would be an *extraordinary* set. Whether there really exist extraordinary sets is open to question, but there certainly exist ordinary ones—indeed, all sets commonly encountered are ordinary. Well, by Frege's principle there exists the set—call it  $\mathcal{O}$ —of all ordinary sets. Thus  $\mathcal{O}$  contains all and only those sets that do not contain themselves (just like the barber who shaves all and only those who do not shave themselves). Is the set  $\mathcal{O}$  ordinary or extraordinary? Suppose  $\mathcal{O}$  is ordinary. Then, since *all* ordinary sets are members of  $\mathcal{O}$ ,  $\mathcal{O}$  must be a member of  $\mathcal{O}$ , which makes  $\mathcal{O}$  extraordinary! This is clearly a contradiction. On the other hand, suppose  $\mathcal{O}$  is extraordinary. Thus  $\mathcal{O}$  is a member of itself, but since only ordinary sets are members of

$\mathcal{O}$ ,  $\mathcal{O}$  must be one of the ordinary sets, which is again a contradiction! Thus there cannot exist a set  $\mathcal{O}$  containing all and only those sets that don't contain themselves (just as there cannot be a barber who shaves all and only those people who don't shave themselves).

Frege's principle is thus seen to lead to a contradiction, hence has to be modified. This has subsequently been done in several ways, which constitutes the field known as *set theory*.

- 12.22. Funny as it may seem at first, the argument *is* valid! Since *everyone* loves my baby, it follows that my baby, being a person, loves my baby. Thus my baby loves my baby, but also loves *only* me. It then follows that my baby is the same person as me!

Of course the argument, though valid, cannot be sound; the premises can't both be true, since they lead to the absurd conclusion that I am my own baby.

## - CHAPTER 13 -

# BEGINNING FIRST-ORDER LOGIC

Now we shall approach first-order logic more systematically.

### INTRODUCING $\forall$ AND $\exists$

In first-order logic we use letters  $x, y, z$ , with or without subscripts, to stand for arbitrary objects of the domain under discussion. What the domain is depends on the application in question. For example, if we are doing algebra, the letters  $x, y, z$  stand for arbitrary *numbers*. If we are doing geometry, the letters  $x, y, z$  might stand for *points* in a plane. If we are doing sociology, the objects in question might be *people*. First-order logic is extremely general and is thus applicable to a wide variety of disciplines.

Given any property  $P$  and any object  $x$ , the proposition that  $x$  has the property  $P$  is neatly symbolized  $Px$ . Now, suppose we wish to say that *every* object  $x$  has property  $P$ ; how do we render this symbolically? Well, here is where we introduce the symbol  $\forall$ —called the *universal quantifier*—which stands for “all,” or “every.” Thus  $\forall x$  is read “for all  $x$ ” or “for every  $x$ ,” and so the English sentence “Everything has the property  $P$ ” is symbolically rendered  $\forall xPx$  (read “For every  $x$ ,  $Px$ ”).

Although in ordinary English, the word “some” *tends* to have a plural connotation, in logic it means only “at least one”; it does not mean “two or more,” only “one or more.” This is important to remember! Now, in

first-order logic, the phrase “there exists at least one” is symbolized by the single symbol  $\exists$ , called the *existential quantifier*. Then the proposition that at least one object  $x$  has the property  $P$  is symbolized  $\exists xPx$ . So first-order logic uses the logical connectives  $\sim$ ,  $\wedge$ ,  $\vee$ ,  $\Rightarrow$  of propositional logic and the quantifiers  $\forall$  and  $\exists$ .

Suppose we now use the letters  $x, y, z$  to stand for unspecified *people*. Let  $Gx$  stand for “ $x$  is good.” Then  $\forall xGx$  says “Everyone is good” and  $\exists xGx$  says “Some people are good” or “At least one person is good” or “There exists a good person.” Now, how do we symbolically say that no one is good? One way to do this is  $\sim\exists xGx$  (it is *not* the case that there exists a good person). An alternative way is  $\forall x(\sim Gx)$  (for every person  $x$ ,  $x$  is not good).

Let us now abbreviate “ $x$  goes to heaven” by  $Hx$ . How do we symbolize “All good people go to heaven?” Well, this can be re-stated: “For every person  $x$ , if  $x$  is good, then  $x$  goes to heaven,” and hence is symbolized  $\forall x(Gx \Rightarrow Hx)$  (for all  $x$ ,  $x$  is good implies  $x$  goes to heaven). What about “*Only* good people go to heaven”? One way to symbolize this is  $\forall x(\sim Gx \Rightarrow \sim Hx)$  (for all people  $x$ , if  $x$  is not good, then  $x$  doesn’t go to heaven). An equivalent rendition is  $\forall x(Hx \Rightarrow Gx)$  (for any person  $x$ , if  $x$  goes to heaven, then  $x$  must be good). What about “Some good people go to heaven”? Remember now, this means only that *at least one* good person goes to heaven, in other words, there exists a good person  $x$  who goes to heaven, or, equivalently, there exists a person  $x$  who is both good and goes to heaven, so the symbolic rendition is simply  $\exists x(Gx \wedge Hx)$ . What about “No good person goes to heaven”? This is simply  $\sim\exists x(Gx \wedge Hx)$ . What about “No good person fails to go to heaven”? This is then  $\sim\exists x(Gx \wedge \sim Hx)$  (there is no person who is good and fails to go to heaven). This is only a roundabout way of saying, however, that all good people go to heaven, so  $\sim\exists x(Gx \wedge \sim Hx)$  is equivalent to  $\forall x(Gx \Rightarrow Hx)$ .

Now let’s consider the old saying “God helps those who help themselves.” Actually, there is a good deal of ambiguity here; does it mean that God helps *all* those who help themselves, or that God helps *only* those who help themselves, or does it mean that God helps *all* those and *only* those who help themselves? Well, let us abbreviate “ $x$  helps  $y$ ” by  $xHy$ , and let “ $g$ ” abbreviate “God.” Then “God helps *all* those who help themselves” would be symbolized  $\forall x(xHx \Rightarrow gHx)$ . What about “God helps *only* those who help themselves”? One rendition is  $\forall x(gHx \Rightarrow xHx)$  (for all  $x$ , God helps  $x$  implies that  $x$  helps  $x$ ). Another is  $\forall x(\sim xHx \Rightarrow \sim gHx)$  (for all  $x$ , if  $x$  doesn’t help  $x$ , then God doesn’t help  $x$ ). Another is  $\sim\exists x(gHx \wedge \sim xHx)$  (there is no person  $x$  such that God helps  $x$  and  $x$  doesn’t help  $x$ ).

Let us consider some more translations.

PROBLEM 13.1. Give symbolic renditions of the following (using  $xHy$  for  $x$  helps  $y$ , and  $g$  for God):

- (a) God helps only those who help God.
- (b) God helps all and only those who help themselves.

PROBLEM 13.2. Let  $h$  stand for Holmes (Sherlock Holmes) and  $m$  for Moriarty. Let us abbreviate " $x$  can trap  $y$ " by  $xTy$ . Give symbolic renditions of the following:

- (a) Holmes can trap anyone who can trap Moriarty.
- (b) Holmes can trap anyone whom Moriarty can trap.
- (c) Holmes can trap anyone who can be trapped by Moriarty.
- (d) If anyone can trap Moriarty, then Holmes can.
- (e) If everyone can trap Moriarty, then Holmes can.
- (f) Anyone who can trap Holmes can trap Moriarty.
- (g) No one can trap Holmes unless he can trap Moriarty.
- (h) Everyone can trap someone who cannot trap Moriarty.
- (i) Anyone who can trap Holmes can trap anyone whom Holmes can trap.

PROBLEM 13.3. Let us abbreviate " $x$  knows  $y$ " by  $xKy$ . Express the following symbolically:

- (a) Everyone knows someone.
- (b) Someone knows everyone.
- (c) Someone is known by everyone.
- (d) Everyone knows someone who doesn't know him.
- (e) There is someone who knows everyone who knows him.

It is interesting to note that the words "anyone" and "anybody" sometimes mean *everyone*, and sometimes mean *someone*. For example, the sentence "Anybody can do it" means that *everybody* can do it, but in the sentence "If anybody can do it, then John can do it" (or "John, if anybody, can do it"), the word "anybody" means *somebody*.

PROBLEM 13.4. Let  $Dx$  abbreviate “ $x$  can do it” and  $j$  abbreviate *John*. Symbolically express the sentence “John, if anybody, can do it.”

PROBLEM 13.5. Let  $Dx$  abbreviate “ $x$  can do it,” let  $j$  abbreviate *John* and let  $x = y$  abbreviate “ $x$  is identical with  $y$ .” How would you express the proposition that John is the *only* one who can do it?

Here are some examples from arithmetic. The letters  $x, y, z$  will now stand for arbitrary natural numbers instead of people. (The natural numbers are  $0, 1, 2, 3, 4, \dots$ , etc.—that is,  $0$  together with the positive whole numbers.) The symbol  $<$  stands for “is less than,” so for any numbers  $x$  and  $y$ ,  $x < y$  is read: “ $x$  is less than  $y$ .”

PROBLEM 13.6. Using the symbol  $<$  and logical connectives and quantifiers, express symbolically the following statements about natural numbers (i.e., where by “number” is meant “natural number”):

- (a)  $x$  is greater than  $y$ .
- (b) For every number there is a greater number.
- (c) For every number there is a lesser number.
- (d) Every number is greater than some number.

Is statement (c) true or false?

### INTERDEPENDENCE OF $\forall$ AND $\exists$

It is possible to define  $\exists$  from  $\forall$  and the logical connectives, and vice versa, as the following two problems indicate.

PROBLEM 13.7. Let  $Gx$  stand for the proposition that  $x$  is good. The statement  $\forall xGx$  says that everyone is good, and  $\exists xGx$  says that at least one person is good. Now, suppose you are living in a country where, for some odd reason, it is illegal to use the symbol  $\forall$ , but you are allowed to use  $\exists$ . You wish to express the proposition that *everyone* is good. How can you do that using only the quantifier  $\exists$  (as well as any of the logical connectives  $\sim, \wedge, \vee, \Rightarrow$ )?

PROBLEM 13.8. In another country you are allowed to use the symbol  $\forall$ , but not the symbol  $\exists$ . How would you then express the proposition that at least one person  $x$  is good?

## FORMULAS, VALIDITY, SATISFIABILITY

### FORMULAS

As the reader already knows, in first-order logic we use the logical connectives  $\sim$ ,  $\wedge$ ,  $\vee$ ,  $\Rightarrow$  and the quantifiers  $\forall$  and  $\exists$ ; and we use small letters  $x$ ,  $y$ ,  $z$ ,  $w$ , with or without subscripts, as variables standing for arbitrary objects of the domain under discussion. These variables are called *individual* variables, and the domain under discussion is called the *range* of the variables. For example, if we are doing number theory, we say that the individual variables *range* over the natural numbers.

We also use capital letters to stand for *properties* of individuals; these letters are called 1-place *predicates*. We also have 2-place predicates standing for (2-place) *relations* between individuals, and 3-place predicates for 3-place relations. (Examples of 3-place relations among individuals  $x$ ,  $y$ ,  $z$  might be, in arithmetic,  $x+y=z$ , or, in geometry, "point  $x$  is *between* points  $y$  and  $z$ ," or if  $x$ ,  $y$ ,  $z$  stand for *people*, we might have a 3-place relation such as " $x$  *introduced*  $y$  to  $z$ ."") More generally, for each positive integer  $n$ , we have letters called  $n$ -place predicates standing for  $n$ -place relations—though, for our purposes, we will not need to go beyond 3-place predicates.

By an *atomic formula* we mean either an expression  $Px$ , where  $x$  is an individual variable and  $P$  is a 1-place predicate, or an expression  $Rxy$  (also written  $xRy$ ) where  $R$  is a 2-place predicate (and  $x$  and  $y$  are individual variables), or, in general, any  $n$ -place predicate followed by  $n$  occurrences of individual variables. Starting with the atomic formulas, one builds the entire class of *first-order formulas* by the following rules: Given formulas  $F$  and  $G$  already constructed, the expressions  $\sim F$ ,  $(F \wedge G)$ ,  $(F \vee G)$ ,  $(F \Rightarrow G)$  are again formulas (as in propositional logic), and  $\forall xG$ ,  $\exists xG$ , where  $x$  is any individual variable, are also formulas. Thus we have the following rules:

- (1) Any atomic formula is a formula.
- (2) For any formulas  $F$  and  $G$ , the expression  $\sim F$ ,  $(F \wedge G)$ ,  $(F \vee G)$ ,  $(F \Rightarrow G)$  are again formulas.
- (3) For any formula  $F$  and any individual variable  $x$ , the expressions  $\forall xF$  and  $\exists xF$  are again formulas.

The formula  $\forall xF$  is called the *universal quantification* of  $F$  with respect to  $x$ , and  $\exists xF$  is called the *existential quantification* of  $F$  with respect to  $x$ .

As in propositional logic, in displaying a formula, we can drop superfluous parentheses if no ambiguity results.

## FREE AND BOUND OCCURRENCES

Before defining these important notions, we look at some examples:

In the arithmetic of the natural numbers, consider the equation

$$x = 5y. \tag{13.1}$$

This equation, as it now stands, is neither true nor false, but becomes so when we assign values to the variables  $x$  and  $y$ . (For example, if we take  $x$  to be 10 and  $y$  to be 2, we have a truth, but if we take  $x$  to be 12 and  $y$  to be 7, we have an obvious falsehood.) The important thing to notice now is that the truth or falsity of (13.1) depends both on a choice of value for  $x$  and a choice for  $y$ . This reflects the purely formal fact that  $x$  and  $y$  occur *freely* in (13.1).

Now consider

$$\exists y(x = 5y). \tag{13.2}$$

The truth or falsity of (13.2) depends on  $x$ , but not on any choice for  $y$ . Indeed, we could restate (13.2) in a form in which the variable  $y$  does not even occur—namely, “ $x$  is divisible by 5.” And this reflects the fact that  $x$  has a free occurrence in (13.2) but  $y$  does not— $y$  is *bound* in (13.2).

Suppose  $x$  has an occurrence in a formula  $F$ . As soon as one puts  $\forall x$  or  $\exists x$  in front of  $F$ , all free occurrences of  $x$  become bound—that is, all occurrences of  $x$  in  $\forall xF$  are bound, and likewise with  $\exists xF$ . Here are the precise rules determining freedom and bondage:

- (1) In an atomic formula, all occurrences of variables are free.
- (2) The free occurrences of a variable in  $\sim F$  are the same as those in  $F$ .  
The free occurrences of a variable in  $F \wedge G$  are those of  $F$  and those of  $G$ . Likewise with  $\vee$  or  $\Rightarrow$  instead of  $\wedge$ .
- (3) All occurrences of a variable  $x$  in  $\forall xF$  are *bound* (not free), but for any variable  $y$  distinct from  $x$ , the free occurrences of  $y$  in  $\forall xF$  are those in  $F$  itself. Similarly with  $\exists$  in place of  $\forall$ .

A formula is called *closed* if no variable is free in it; otherwise it is called *open*.

## INTERPRETATIONS

An *interpretation* of a formula is given by first specifying the domain of individuals, then assigning to each predicate a property of, or a relation among, the individuals, and, lastly, if there are free variables in the formula, assigning an individual to each of the free variables. Once an interpretation is given, the formula then becomes either true or false under the interpretation.

Let us consider some examples: Take the formula  $\exists xRxy$ . In this formula,  $x$  is bound and  $y$  is free. For one interpretation, let the domain of individuals be the set of all people who have ever lived in this world. Define  $Rxy$  to be the relation “ $x$  is the parent of  $y$ .” If we take for  $y$  the person Abraham Lincoln, then the formula is certainly true under that interpretation (Lincoln certainly had a parent), but if we take  $y$  to be Adam or Eve, the sentence is then probably false.

Suppose we alternatively define  $Rxy$  to mean that  $x$  is *married* to  $y$ . Then,  $\exists xRxy$  simply says that  $y$  is married, which is true for some  $y$ 's and false for others. Thus  $\exists y\exists xRxy$  is true under this interpretation, but  $\forall y\exists xRxy$  is false (it is false that everyone is married).

Let us now consider another interpretation of this same formula—an interpretation in the domain of natural numbers. A number  $x$  is said to *properly* divide  $y$  if  $x$  divides  $y$  but  $x \neq y$  and  $x \neq 1$ . Now let us interpret  $Rxy$  to mean that  $x$  properly divides  $y$ . Then  $\exists xRxy$  is true for some  $y$ 's and false for others (for example, it is true for  $y=12$  but false for  $y=7$ ). A number is called *prime* if no number properly divides it and *composite* otherwise. Under our present interpretation,  $\exists xRxy$  simply says that  $y$  is composite (and thus the formula  $\sim\exists xRxy$  says that  $y$  is prime).

### VALIDITY AND SATISFIABILITY

A formula is called *valid in a given domain* if it is true under *every* interpretation in that domain, and is called *logically valid*—or just *valid*, for short—if it is valid in all domains that contain at least one element. Thus a valid formula is one that is true under all possible interpretations in all possible domains, except perhaps a domain that contains no elements.

A word about sets that contain no elements—so-called empty sets. This notion may seem strange at first, but a good example of an empty set is the set of all people in a theater after everyone has left. Another example is the set of all even prime numbers greater than 2. There simply are no such numbers, because any even number greater than 2 is properly divisible by 2, so the set of all even primes greater than 2 is empty.

Thus a formula is called *valid* if it is valid in all *non-empty* domains. Now, some curious things happen with interpretations in an *empty* domain! We recall that if a certain club contains no Frenchmen, then the statement that *all* Frenchmen in the club wear hats is true. *Anything* one says about *all* Frenchmen in the club is true if there are no Frenchmen in the club. In general, whatever one says about *all* members of an empty domain is true—the formula  $\forall xPx$  is true for an empty domain, regardless of what the property  $P$  is. The formula  $\forall xPx$  is thus *valid in an empty domain*—though it is of course not valid in any non-empty domain, and

hence is not valid. Thus  $\forall xPx$  is an example of an invalid formula that is nevertheless valid in an empty domain.

A formula is said to be *satisfiable in a given domain* if it is true under at least one interpretation in the domain, and it is called *satisfiable*, period, if it is satisfiable in at least one *non-empty* domain. Let us note that to say that a formula  $F$  is not valid is equivalent to saying that its negation  $\sim F$  is satisfiable, and that to say that  $F$  is valid is to say that  $\sim F$  is not satisfiable. Also  $F$  is satisfiable if and only if  $\sim F$  is not valid, and  $F$  is unsatisfiable if and only if  $\sim F$  is valid.

We have seen that the formula  $\forall xPx$  is valid in an empty domain but not valid in any non-empty domain. Is there a formula that is valid in all non-empty domains but not valid in an empty domain? Yes, there is. Better yet, there is a formula that is valid in all non-empty domains but is not only not valid in an empty domain, but not even satisfiable in an empty domain.

PROBLEM 13.9. Find such a formula.

If the reader is in the mood for trying a really difficult one, here it is:

PROBLEM 13.10. Find a formula that is not satisfiable in any *finite* non-empty domain, but is satisfiable in some infinite domains.

## TAUTOLOGIES

A formula  $X$  of first-order logic is said to be an *instance* of a formula  $Y$  of propositional logic if  $X$  is obtainable from  $Y$  by substituting formulas of first-order logic for the propositional variables in  $Y$ . For example,  $\exists xQx \vee \sim \forall yPy$  is an instance of  $p \vee \sim q$  (it is obtainable by substituting  $\exists xQx$  for  $p$  and  $\forall yPy$  for  $q$ ). Now,  $X$  is called a *tautology* if it is an instance of a tautology of propositional logic. For example,  $\forall xQx \vee \sim \forall xQx$  is a tautology, because it is an instance of the propositional tautology  $p \vee \sim p$ . Even if one did not know what the symbol  $\forall$  meant, one would know that *whatever* it meant,  $\forall xQx \vee \sim \forall xQx$  must be true, because for *any* proposition, either it or its negation must be true. Thus the truth of  $\forall xQx \vee \sim \forall xQx$  is obtainable just from propositional logic; it can be shown by a truth table, taking  $\forall xQx$  as a unit. Another tautology is  $(\forall xPx \wedge \forall xQx) \Rightarrow \forall xPx$ —it is of the form (an instance of)  $(p \wedge q) \Rightarrow p$ . Now, the formula  $(\forall xPx \wedge \forall xQx) \Rightarrow \forall x(Px \wedge Qx)$ , though valid, is *not* a tautology! It is valid, because if *every*  $x$  has property  $P$  and *every*  $x$  has property  $Q$ , then of course every  $x$  has both properties  $P$  and  $Q$ . But it is not an instance of any tautology of propositional logic. Also, to realize the validity of the formula, one must know what the symbol  $\forall$  means—for example, if one reinterpreted  $\forall$  to mean “there exists,” instead of “for all,” the formula wouldn’t always be true (if some element has the property  $P$  and

some element has the property  $Q$ , it doesn't necessarily follow that some single element has both properties  $P$  and  $Q$ ). All tautologies are of course valid, but they constitute only a fragment of the class of valid formulas! It is the valid formulas that we are now interested in, and we seek a general proof procedure for valid formulas. Such a procedure is the method of *tableaux*, to which we now turn.

## TABLEAUX FOR FIRST-ORDER LOGIC

Tableaux for formulas of first-order logic use the eight rules of propositional logic together with four new rules for the quantifiers, which we will shortly present. First for some examples.

Suppose we wish to prove the formula  $\exists xPx \Rightarrow \sim \forall x \sim (Px \vee Qx)$ . As with propositional logic, we start the tableaux with  $F$  followed by the formula we wish to prove.

$$(1) \quad F\exists xPx \Rightarrow \sim \forall x \sim (Px \vee Qx)$$

Next, we use rules from propositional logic to extend the tableaux thus. (Remember, we put to the right of each line the number of the line from which it was inferred.)

$$(2) \quad T\exists xPx \quad (1)$$

$$(3) \quad F\sim \forall x \sim (Px \vee Qx) \quad (1)$$

$$(4) \quad T\forall x \sim (Px \vee Qx) \quad (3)$$

There is no rule from propositional logic that is now applicable, so we start working on the quantifiers. By (2), there is at least one  $x$  such that  $Px$ ; let  $a$  be the name of any such  $x$ . So we add

$$(5) \quad TP_a \quad (2)$$

Next we look at line (4), which says that whatever  $x$  we take, it is not the case that  $Px$  or  $Qx$ ; in particular, it is not the case that  $Pa$  or  $Qa$ , so we add:

$$(6) \quad T\sim(P_a \vee Q_a) \quad (4)$$

At this point, (5) and (6) constitute a clear inconsistency in propositional logic, and we can stop right here, or alternatively close the tableau using only tableau rules of propositional logic thus:

$$(7) \quad FPa \vee Qa \quad (6)$$

$$(8) \quad \underline{FP_a} \quad (7)$$

(Line 8 clashes with line 5.)

In the future, to save unnecessary work, whenever we obtain a branch that contains an inconsistency in propositional logic, we can put a bar under it and treat it as closed—since we know it *can be* closed (in fact, closed using only tableau rules of propositional logic).

Let us consider another example: We wish to prove the formula

$$(\forall x Px \wedge \exists x (Px \Rightarrow Qx)) \Rightarrow \exists x Qx.$$

Here is a tableau that does this (explanations follow):

|                           |                                                                                 |                            |
|---------------------------|---------------------------------------------------------------------------------|----------------------------|
| (1)                       | $F(\forall x Px \wedge \exists x (Px \Rightarrow Qx)) \Rightarrow \exists x Qx$ |                            |
| (2)                       | $T\forall x Px \wedge \exists x (Px \Rightarrow Qx)$                            | (1)                        |
| (3)                       | $F\exists x Qx$                                                                 | (1)                        |
| (4)                       | $T\forall x Px$                                                                 | (2)                        |
| (5)                       | $T\exists x (Px \Rightarrow Qx)$                                                | (2)                        |
| (6)                       | $TPa \Rightarrow Qa$                                                            | (5)                        |
| (7)                       | $TPa$                                                                           | (4)                        |
| (8)                       | $FQa$                                                                           | (3)                        |
| $\swarrow \quad \searrow$ |                                                                                 |                            |
| (9)                       | $\underline{F}Pa$ (6)                                                           | (10) $\underline{T}Qa$ (6) |

EXPLANATIONS. Lines (2), (3), (4), (5) were obtained only by tableau rules of propositional logic. Now, line (5) says that  $Px \Rightarrow Qx$  holds for at least one  $x$ , and so we let  $a$  be such an  $x$ , and thus get line (6). Line (4) says that  $Px$  holds for every  $x$ , and so in particular,  $Pa$  holds, which gives us line (7). Line (3) says that it is false that there is any  $x$  such that  $Qx$ , so, in particular,  $Qa$  is false, which gives us line (8). Then, of course, line (6) branches to (9) and (10), and the tableau then closes.

### TABLEAU RULES FOR FIRST-ORDER LOGIC

It is technically convenient to have a group of letters called *parameters*, which we use to substitute for free occurrences of variables in formulas. We shall use the letters  $a, b, c$ , with or without subscripts, as parameters. For any formula  $G$  and any variable  $x$  and any parameter  $a$ , by  $G_x(a)$  we shall mean the result of substituting  $a$  for every *free* occurrence of  $x$  in  $G$ . Thus:

- (1) If  $G$  is an *atomic* formula, then  $G_x(a)$  is the result of substituting  $a$  for every occurrence of  $x$  in  $G$ . (Of course, if  $x$  does not occur in  $G$ , then  $G_x(a)$  is  $G$  itself.)
- (2) For any formula, the formula  $(\sim G)_x(a)$  is  $\sim G_x(a)$ —that is, to substitute  $a$  for all free occurrences of  $x$  in  $\sim G$ , we first substitute  $a$  for all free occurrences of  $x$  in  $G$ , thus obtaining  $G_x(a)$ , and then take its negation  $\sim G_x(a)$ .

- (3) For any formulas  $G$  and  $H$ , the formula  $(G \wedge H)_x(a)$  is  $G_x(a) \wedge H_x(a)$ . (That is, to substitute  $a$  for all free occurrences of  $x$  in  $G \wedge H$ , we first substitute  $a$  for all free occurrences of  $x$  in  $G$ , obtaining  $G_x(a)$ , then do the same thing with  $H$ , obtaining  $H_x(a)$ , and then take the conjunction  $G_x(a) \wedge H_x(a)$ .) Similarly,

$$(G \vee H)_x(a) = G_x(a) \vee H_x(a),$$

$$(G \Rightarrow H)_x(a) = G_x(a) \Rightarrow H_x(a),$$

$$(G \equiv H)_x(a) = G_x(a) \equiv H_x(a).$$

- (4) (a) For any variable  $y$  distinct from  $x$ , the formula  $(\forall y G)_x(a)$  is  $\forall y G_x(a)$ . (That is, the result of substituting  $a$  for all free occurrences of  $x$  in  $\forall y G$  consists of the universal quantification  $\forall y$  followed by the result  $G_x(a)$  of substituting  $a$  for all free occurrences of  $x$  in  $G$ .)
- (b) But for the variable  $x$  itself,  $(\forall x G)_x(a)$  is simply  $\forall x G$ , since  $x$  has no free occurrence in  $\forall x G$ .
- (5) Similarly,  $(\exists y G)_x(a) = \exists y G_x(a)$  if  $y \neq x$ , but  $(\exists x G)_x(a)$  is simply  $\exists x G$ .

A common, and useful, notational convention is to let  $\phi(x)$  be a formula in which the variable  $x$  is the one to be acted on; then take  $\phi(a)$  to be the result of substituting  $a$  for all free occurrences of  $x$ . (Thus if  $G$  is the formula  $\phi(x)$ , then  $\phi(a)$  is  $G_x(a)$ .)

### THE TABLEAU RULES FOR THE QUANTIFIERS

We have four quantifier rules—one for each of  $T\forall x\phi(x)$ ,  $F\forall x\phi(x)$ ,  $T\exists x\phi(x)$ ,  $F\exists x\phi(x)$ . None of the rules are branching.

**RULE  $T\forall$ .** From  $T\forall x\phi(x)$ , we may directly infer  $T\phi(a)$ , where  $a$  is any parameter.

**RULE  $T\exists$ .** This is more delicate! From  $T\exists x\phi(x)$  we may directly infer  $T\phi(a)$ , *provided  $a$  is a parameter that has not been used before!*

Here is the reason for the proviso: Suppose in the course of a proof we show that there is some  $x$  having property  $P$ ; we then can say: "Let  $a$  be such an  $x$ ." Now, suppose we later show that there is some  $x$  having property  $Q$ ; we cannot legitimately say: "Let  $a$  be such an  $x$ ," because we have already committed the symbol  $a$  to being the name of some  $x$  having property  $P$ , and we do not know that there is some  $x$  having *both* property  $P$  and property  $Q$ ! So we must take a new symbol  $b$ , and say: "Let  $b$  be an  $x$  having property  $Q$ ."

**RULE  $F\forall$ .** This is similar to Rule  $T\exists$ : From  $F\forall x\phi(x)$  we may directly infer  $F\phi(a)$ , again *provided that  $a$  is new!*

Here's why:  $F\forall x\phi(x)$  says that it is false that  $\phi(x)$  holds for every  $x$ , which is equivalent to saying that there is at least one  $x$  such that  $\phi(x)$  is false; we let  $a$  be such an  $x$ , and so write  $F\phi(a)$ . Again,  $a$  must be a new parameter for the same reasons that we explained in Rule  $T\exists$ .

(We might remark that instead of Rule  $F\forall$ , we could just as well have taken as a rule the following: "From  $F\forall x\phi(x)$  we can directly infer  $T\exists x\sim\phi(x)$ ." But then it would take three steps to get  $F\phi(a)$  from  $F\forall x\phi(x)$ —namely  $F\forall x\phi(x)$ ,  $T\exists x\sim\phi(x)$ ,  $T\sim\phi(a)$  (by Rule  $T\exists$ ), and then  $F\phi(a)$ .)

**RULE  $F\exists$ .** From  $F\exists x\phi(x)$  we may infer  $F\phi(a)$  for *any* parameter  $a$  (no restriction necessary).

You see,  $F\exists x\phi(x)$  says that it is false that there is *any*  $x$  such that  $\phi(x)$ —in other words,  $\phi(x)$  is false for *every*  $x$ , and so in particular,  $\phi(a)$  is false, and thus  $F\phi(a)$ . (Again, this rule could be replaced by the rule "From  $F\exists x\phi(x)$  we can infer  $T\forall x\sim\phi(x)$ .")

Let us now review the four rules in schematic form:

|                                    |                                                                               |                                    |                                                                               |
|------------------------------------|-------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------|
| <b>RULE <math>T\forall</math>.</b> | $\frac{T\forall x\phi(x)}{T\phi(a)}$ <p>(<math>a</math> is any parameter)</p> | <b>RULE <math>F\exists</math>.</b> | $\frac{F\exists x\phi(x)}{F\phi(a)}$ <p>(<math>a</math> is any parameter)</p> |
| <b>RULE <math>T\exists</math>.</b> | $\frac{T\exists x\phi(x)}{T\phi(a)}$ <p>(<math>a</math> must be new)</p>      | <b>RULE <math>F\forall</math>.</b> | $\frac{F\forall x\phi(x)}{F\phi(a)}$ <p>(<math>a</math> must be new)</p>      |

Rules  $T\forall$  and  $F\exists$  are collectively called *universal* rules ( $F\exists x\phi(x)$  asserts the *universal* fact that for *every* element  $a$ , it is false that  $\phi(a)$ ). The rules  $T\exists$  and  $F\forall$  are called *existential* rules ( $F\forall x\phi(x)$  asserts the *existential* fact that there exists at least one element  $a$  such that  $\phi(a)$  is false).

For *unsigned* formulas, the quantificational tableau rules are

|                                   |                                                                             |                                       |                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------|
| <b>RULE <math>\forall</math>.</b> | $\frac{\forall x\phi(x)}{\phi(a)}$ <p>(<math>a</math> is any parameter)</p> | <b>RULE <math>\sim\exists</math>.</b> | $\frac{\sim\exists\phi(x)}{\sim\phi(a)}$ <p>(<math>a</math> is any parameter)</p> |
| <b>RULE <math>\exists</math>.</b> | $\frac{\exists x\phi(x)}{\phi(a)}$ <p>(<math>a</math> must be new)</p>      | <b>RULE <math>\sim\forall</math>.</b> | $\frac{\sim\forall\phi(x)}{\sim\phi(a)}$ <p>(<math>a</math> must be new)</p>      |

## UNIFIED NOTATION

We recall the unifying  $\alpha$ ,  $\beta$  notation for propositional logic. It will save us considerable circumlocution to extend this now to first-order logic.

We continue to use  $\alpha$  and  $\beta$  the same way as we did in propositional logic, except that “formula” is now construed as a *closed* formula of first-order logic. We now add two more categories,  $\gamma$  and  $\delta$ , as follows.

$\gamma$  shall be any formula of the universal type— $T\forall x\phi(x)$  or  $F\exists x\phi(x)$ —and, for any parameter  $a$ , by  $\gamma(a)$  we shall mean  $T\phi(a)$ ,  $F\phi(a)$ , respectively. We refer to any such formula  $\gamma(a)$  as a *component* of  $\gamma$ .  $\delta$  shall be any formula of the *existential* type—either  $T\exists x\phi(x)$  or  $F\forall x\phi(x)$ —and by  $\delta(a)$  we shall respectively mean  $T\phi(a)$ ,  $F\phi(a)$ . We refer to any such formula  $\delta(a)$  as a *component* of  $\delta$ . Our universal rules  $T\forall$  and  $F\exists$  are now both subsumed under Rule C below, and our existential rules are both subsumed under Rule D below:

|                                                                                            |                                                                                           |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <p>RULE C. <math>\frac{\gamma}{\gamma(a)}</math><br/>(<math>a</math> is any parameter)</p> | <p>RULE D. <math>\frac{\delta}{\delta(a)}</math><br/>(provided <math>a</math> is new)</p> |
|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|

And, to review the propositional rules,

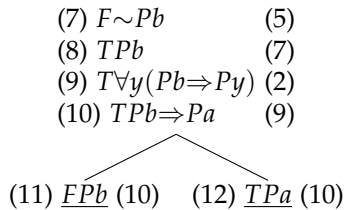
|                                                                    |                                                                 |
|--------------------------------------------------------------------|-----------------------------------------------------------------|
| <p>RULE A. <math>\frac{\alpha}{\alpha_1 \quad \alpha_2}</math></p> | <p>RULE B. <math>\frac{\beta}{\beta_1 \quad \beta_2}</math></p> |
|--------------------------------------------------------------------|-----------------------------------------------------------------|

Thus, using our unifying notation, our 12 tableau rules for first-order logic are collapsed into four. This completes the list of tableau rules for first-order logic. We note that Rule B is the only branching rule; all the quantificational rules for first-order logic are direct.

For *unsigned* formulas, we let  $\gamma$  be any formula of the form  $\forall x\phi(x)$  or  $\sim\exists x\phi(x)$ , and by  $\gamma(a)$  we respectively mean  $\phi(a)$ ,  $\sim\phi(a)$ . We let  $\delta$  be any formula of the form  $\exists x\phi(x)$  or  $\sim\forall x\phi(x)$ , and by  $\delta(a)$  we respectively mean  $\phi(a)$ ,  $\sim\phi(a)$ .

Let us now try another tableau—let us prove the formula  $\forall x\forall y(Px \Rightarrow Py) \Rightarrow (\forall xPx \vee \forall x\sim Px)$ :

- |                                                                                              |     |
|----------------------------------------------------------------------------------------------|-----|
| (1) $F\forall x\forall y(Px \Rightarrow Py) \Rightarrow (\forall xPx \vee \forall x\sim Px)$ |     |
| (2) $T\forall x\forall y(Px \Rightarrow Py)$                                                 | (1) |
| (3) $F\forall xPx \vee \forall x\sim Px$                                                     | (1) |
| (4) $F\forall xPx$                                                                           | (3) |
| (5) $F\forall x\sim Px$                                                                      | (3) |
| (6) $FPa$                                                                                    | (4) |



DISCUSSION. In line (7), in accordance with Rule D, I couldn't use the parameter  $a$  a second time, so I had to take a new parameter  $b$ . Now, in line (9), how did I know that I had best use the parameter  $b$  rather than  $a$ , or some other parameter? I knew because I informally went through a proof in my mind before I made the tableau, and then made the tableau accordingly. This brings us to some important points.

Tableaux for propositional logic are purely routine things; it makes no essential difference in which order one uses the rules; if one order leads to a closure, then any other order will, also. But with first-order tableaux, the situation is entirely different: For one thing, in constructing a tableau in propositional logic, if one does not repeat any formula, then the tableau must terminate in a finite number of steps; in a tableau for first-order logic, the process can go on indefinitely, because when we use a universal formula  $\gamma$ , we can infer  $\gamma(a)$ ,  $\gamma(b)$ ,... with no limit to the number of parameters we can use. I might remark that an existential formula  $\delta$  need be used only once; one is *allowed* to use it more than once, but there is never any advantage to doing so; with a  $\gamma$ , we have no idea in advance how many parameters have to be used to effect a closure. Moreover, if one doesn't do things in the right order, the tableau might run on forever without ever closing, even though the tableau could be closed by proceeding differently. Isn't there some systematic procedure that, if followed, will guarantee closure, if closure is possible? Yes, there is, and we will consider one in a later chapter. Following such a procedure is purely mechanical, and it is easy to program a computer to do so. A tableau constructed with intelligence and creative ingenuity, however, often closes much more quickly than one constructed by following the mechanical procedure! (In general, mechanical procedures are unimaginative and incapable of seizing clever strategies.)

In using the universal rule (Rule C), it is best to use parameters that are already on the tree, rather than new ones. Which parameters to use is where ingenuity comes in!

When proving a biconditional  $X \equiv Y$ , it produces less clutter to make one tableau starting with  $TX$  and  $FY$  and another starting with  $FX$  and  $TY$ .

EXERCISE 13.1. By tableaux, prove the following formulas:

- (a)  $\forall y(\forall xPx \Rightarrow Py)$ .
- (b)  $\forall x(Px \Rightarrow \exists xPx)$ .
- (c)  $\sim \exists yPy \Rightarrow \forall y(\exists xPx \Rightarrow Py)$ .
- (d)  $\exists xPx \Rightarrow \exists yPy$ .
- (e)  $(\forall xPx \wedge \forall xQx) \equiv \forall x(Px \wedge Qx)$ .
- (f)  $(\forall xPx \vee \forall xQx) \Rightarrow \forall x(Px \vee Qx)$ .

The converse,  $\forall x(Px \vee Qx) \Rightarrow (\forall xPx \vee \forall xQx)$ , is not valid. Why?

- (g)  $\exists x(Px \vee Qx) \equiv (\exists xPx \vee \exists xQx)$ .
- (h)  $\exists x(Px \wedge Qx) \Rightarrow (\exists xPx \wedge \exists xQx)$ .

The converse is not valid. Why?

EXERCISE 13.2. In the next group,  $C$  is any formula in which  $x$  does not occur free (and hence for any parameter  $a$ , the formula  $C_x(a)$  is  $C$  itself).

- (a)  $\forall x(Px \vee C) \equiv (\forall xPx \vee C)$ .
- (b)  $\exists x(Px \wedge C) \equiv (\exists xPx \wedge C)$ .
- (c)  $\exists xC \equiv C$ .
- (d)  $\forall xC \equiv C$ .
- (e)  $\exists x(C \Rightarrow Px) \equiv (C \Rightarrow \exists xPx)$ .
- (f)  $\exists x(Px \Rightarrow C) \equiv (\forall xPx \Rightarrow C)$ .
- (g)  $\forall x(C \Rightarrow Px) \equiv (C \Rightarrow \forall xPx)$ .
- (h)  $\forall x(Px \Rightarrow C) \equiv (\exists xPx \Rightarrow C)$ .
- (i)  $\forall x(Px \equiv C) \Rightarrow (\forall xPx \vee \forall x \sim Px)$ .

### THE DRINKING FORMULA

Here is a formula whose validity comes as a surprise to most people. In presenting this formula to a graduate logic class, I prefaced it with the following joke:

A man at a bar suddenly slammed down his fist and said: "Gimme a drink, and give everyone in the house a drink, because when I drink, *everyone* drinks!" So drinks were happily passed around the house. A few minutes later, the man said: "Gimme another drink, and give everyone else a drink, because when I take another drink, *everyone* takes another drink." So, second drinks were happily passed around the house. Then the man slammed some money on the counter and said: "And when I pay, *everybody* pays!"

This concludes the joke. The interesting thing is that there really must be a person  $x$  such that if  $x$  drinks, then *everybody* drinks! [ $\exists x(Dx \Rightarrow \forall yDy)$ ,  $Dy$  meaning that  $y$  drinks.] Before we prove this by a tableau, consider the following informal argument: Either everybody drinks or not everybody drinks. Suppose everybody drinks:  $\forall yDy$ . Since  $\forall yDy$  is true, it is implied by any proposition; hence for *any*  $x$ ,  $Dx \Rightarrow \forall yDy$  holds, so of course there is some  $x$  such that  $Dx \Rightarrow \forall yDy$  (in fact any  $x$  will do). Now consider the case when not everyone drinks. Then there is at least one person—call him Jim—who doesn't drink. Let  $j$  abbreviate "Jim." Thus  $Dj$  is false, and since a false proposition implies every proposition, it follows that  $Dj \Rightarrow \forall yDy$  is true! And so there does exist an  $x$ —namely  $j$ —such that  $Dx \Rightarrow \forall yDy$ .

Now let's prove the formula  $\exists x(Dx \Rightarrow \forall yDy)$  by a tableau.

- |                                              |                               |
|----------------------------------------------|-------------------------------|
| (1) $F\exists x(Dx \Rightarrow \forall yDy)$ |                               |
| (2) $FDa \Rightarrow \forall yDy$            | (1)                           |
| (3) $TDa$                                    | (2)                           |
| (4) $F\forall yDy$                           | (2)                           |
| (5) $FDb$                                    | (4) [We cannot write $FDa$ .] |
| (6) $FDb \Rightarrow \forall yDy$            | (1)                           |
| (7) $TDb$                                    | (6)                           |

(My graduate students have dubbed this formula the *drinking* formula!)

A more natural proof of the above formula is possible if we use a *synthetic* tableau, as we will now see.

### SYNTHETIC TABLEAUX

As with propositional logic, we define a *synthetic tableau* as one using the splitting rule (at any stage we can take any formula  $X$  and let any branch split to the two formulas  $TX$ ,  $FX$ ).

With a synthetic tableau, we can obtain a proof of the drinking formula that comes closer to the informal argument previously given. Recall

that, in that informal argument, we split the argument into two cases— $\forall xDx$  is true,  $\forall xDx$  is false. We carry that idea into the tableau, which begins thus:

$$\begin{array}{c} (1) F\exists x(Dx \Rightarrow \forall yDy) \\ \swarrow \quad \searrow \\ (2) T\forall yDy \quad (3) F\forall yDy \end{array}$$

We have thus split right after (1). Now we work on the right branch and continue the tableau thus:

$$\begin{array}{c} (1) F\exists x(Dx \Rightarrow \forall yDy) \\ \swarrow \quad \searrow \\ (2) T\forall yDy \quad (3) F\forall yDy \\ \quad \quad \quad (4) FDa \quad (3) \end{array}$$

We are free to use the parameter  $a$  again in both (2) and (3), so we finish the tableau thus:

$$\begin{array}{c} (1) F\exists x(Dx \Rightarrow \forall yDy) \\ \swarrow \quad \searrow \\ \begin{array}{ll} (2) T\forall yDy & (3) F\forall yDy \\ (5) TDa \quad (2) & (4) FDa \quad (3) \\ (6) FDa \Rightarrow \forall yDy \quad (1) & (8) FDa \Rightarrow \forall yDy \quad (1) \\ (7) \underline{F\forall yDy} \quad (6) & (9) \underline{TDa} \quad (8) \end{array} \end{array}$$

We might note that in the analytic tableau proof of the formula, we had to use two parameters, whereas in the synthetic tableau proof we need only one.

**PROBLEM 13.11.** Is it necessarily true that there exists a person  $x$  such that if anybody drinks, then  $x$  drinks?

**PROBLEM 13.12 (ANOTHER CURIOUS ONE).** Let us define a *lover* as anyone who loves at least one person. Now, suppose we are given the following two facts:

- (1) Everyone loves a lover.
- (2) John loves Mary.

Does it logically follow from (1) and (2) that Iago loves Othello?

## LOGICAL CONSEQUENCE

We say that a formula  $Y$  is a *logical consequence* of a formula  $X$  if  $Y$  is true in all interpretations (in a non-empty domain) in which  $X$  is true, or, what is the same thing, if  $X \Rightarrow Y$  is logically valid. To prove that  $Y$  is a logical consequence of  $X$ , it suffices to construct a closed tableau starting with  $TX, FY$ .

We say that  $Z$  is a logical consequence of  $X$  and  $Y$  if  $Z$  is true in all interpretations (in a non-empty domain) in which  $X$  and  $Y$  are both true or, equivalently, if  $(X \wedge Y) \Rightarrow Z$  is logically valid—and a closed tableau starting with  $TX, TY$  and  $FZ$  will establish this, and similarly with four or more formulas.

PROBLEM 13.13. Consider the following argument:

Everyone has a mother.  
 Anybody's mother is that person's parent.  
 —————  
 $\therefore$  Everyone has a parent.

Symbolize the above argument, letting  $Mxy$  mean that  $x$  is the mother of  $y$ , and  $Pxy$  mean that  $x$  is the parent of  $y$ .

EXERCISE 13.3. After symbolizing the above argument, show by a tableau that the conclusion is really a logical consequence of the two premises.

## KNIGHTS AND KNAVES REVISITED

Let us now look at some of the problems of Chapter 12 from the viewpoint of first-order logic.

ALL OR NOTHING. We are on an island where each inhabitant is either a knight or knave. For any inhabitant  $x$ , we let  $Kx$  be the proposition that  $x$  is a knight. Then  $\sim Kx$  says that  $x$  is a knave. Whenever an inhabitant  $x$  asserts a proposition  $\mathcal{P}$ , we know that if  $x$  is a knight then  $\mathcal{P}$  is true, and if  $x$  is a knave then  $\mathcal{P}$  is false—in other words,  $x$  is a knight if and only if  $\mathcal{P}$  is true. Thus we translate “ $x$  asserts  $\mathcal{P}$ ” as  $Kx \equiv \mathcal{P}$ . Now, in Problem 12.1, each inhabitant asserted that all the inhabitants were of the same type, all knights or all knaves. Thus each inhabitant  $x$  asserted  $\forall x Kx \vee \forall x \sim Kx$ , so for each  $x$  we have  $Kx \equiv (\forall x Kx \vee \forall x \sim Kx)$ . Since this holds for each  $x$ , we have  $\forall x (Kx \equiv (\forall x Kx \vee \forall x \sim Kx))$ . We saw in the solution that  $\forall x Kx$  must hold (all the inhabitants are knights). The essence of this problem is that the following formula is logically valid, and can be proved by a tableau:

$$\forall x (Kx \equiv (\forall x Kx \vee \forall x \sim Kx)) \Rightarrow \forall x Kx.$$

EXERCISE 13.4. By a tableau, prove the above formula.

Let us now compare this problem with Problem 12.15, which was that certain inhabitants of earth were classified as *special* or *non-special* and that for each inhabitant  $x$ ,  $x$  is special if and only if either everyone is special or no one is special. Letting  $Sx$  mean that  $x$  is special, we thus have  $\forall x(Sx \equiv (\forall x Sx \vee \forall x \sim Sx))$ , and the solution concludes that  $\forall x Sx$  (everyone is special), so the logical content of the problem is that the formula  $\forall x(Sx \equiv (\forall x Sx \vee \forall x \sim Sx)) \Rightarrow \forall x Sx$  is logically valid. But this is the same as the formula a few lines back, except that “ $S$ ” has replaced “ $K$ ”! Thus, as stated in the solutions in Chapter 12, the two problems are really the same, and the reader can now see why.

SMOKING KNAVES. In Problem 12.2, each inhabitant said: “Some of us are knights and some of us are knaves.” Thus each  $x$  asserted  $\exists x Kx \wedge \exists x \sim Kx$ , so the reality is  $\forall x(Kx \equiv (\exists x Kx \wedge \exists x \sim Kx))$ . The conclusion was  $\forall x \sim Kx$  (all are knaves), therefore  $\forall x \sim Kx$  is a logical consequence of  $\forall x(Kx \equiv (\exists x Kx \wedge \exists x \sim Kx))$ .

EXERCISE 13.5. Prove this by a tableau.

It should now be obvious to the reader that Problem 12.16 is really the same thing.

SMOKING KNIGHTS. Let us now look at Problem 12.4. Each inhabitant asserted that all the knights smoke. Thus  $\forall x(Kx \equiv \forall x(Kx \Rightarrow Sx))$  (where  $Sx$  means that  $x$  smokes). The conclusion was that all are knights and all smoke ( $\forall x(Kx \wedge Sx)$ ).

EXERCISE 13.6. Construct a closed tableau starting with  $T\forall x(Kx \equiv \forall x(Kx \Rightarrow Sx))$  and  $F\forall x(Kx \wedge Sx)$ .

NONSMOKING KNAVES. The logical content of Problem 12.5 is that the statement  $\forall x(Kx \wedge \sim Sx)$  is a logical consequence of  $\forall x(Kx \equiv \exists x(Sx \wedge \sim Kx))$ .

EXERCISE 13.7. Prove this with a tableau.

We close this chapter leaving two vital questions up in the air:

- (1) Can more formulas be proved with synthetic tableaux than with analytic tableaux?
- (2) Can all valid formulas be proved with synthetic tableaux? Can they all be proved with analytic tableaux?

The answers to these questions constitute *major* discoveries in the field of mathematical logic! We will see the answers in a later chapter, but we must first turn to the subject of infinity and mathematical induction. We will take temporary leave of the study of first-order logic and return to it later, after we have laid the adequate groundwork.

## SOLUTIONS

- 13.1. (a)  $\forall x(gHx \Rightarrow xHg)$ .  
Alternatively,  $\forall x(\sim xHg \Rightarrow \sim gHx)$ .  
(b)  $\forall x(gHx \equiv xHx)$ .
- 13.2. (a)  $\forall x(xTm \Rightarrow hTx)$ .  
(b)  $\forall x(mTx \Rightarrow hTx)$ .  
(c) Same as (b).  
(d)  $\exists x(xTM) \Rightarrow hTm$ .  
(e)  $\forall x(xTm) \Rightarrow hTm$ .  
(f)  $\forall x(xTh \Rightarrow xTm)$ .  
(g) Same as (f)!  
(h)  $\forall x \exists y(xTy \wedge \sim yTm)$ .  
(i)  $\forall x(xTh \Rightarrow \forall y(hTy \Rightarrow xTy))$ .  
Alternatively,  $\forall x \forall y((xTh \wedge hTy) \Rightarrow xTy)$ . (This is a bit simpler.)
- 13.3. (a)  $\forall x \exists y xKy$ .  
(b)  $\exists x \forall y xKy$ .  
(c)  $\exists x \forall y yKx$ .  
(d)  $\forall x \exists y(xKy \wedge \sim yKx)$ .  
(e)  $\exists x \forall y(yKx \Rightarrow xKy)$ .
- 13.4.  $\exists x Dx \Rightarrow Dj$ .  
Alternatively,  $\forall x(Dx \Rightarrow Dj)$ .
- 13.5. This is a bit tricky! A solution is  $Dj \wedge \forall x(Dx \Rightarrow x=j)$ .  
An alternative solution is  $\forall x(Dx \equiv x=j)$ .
- 13.6. (a)  $y < x$  (this says that  $y$  is less than  $x$ , which is equivalent to saying that  $x$  is greater than  $y$ ).  
(b)  $\forall x \exists y(x < y)$ .  
(c)  $\forall x \exists y(y < x)$ .  
(d) Same as (c).

Statement (c) is false, of course; no natural number is less than zero.  
But false statements can be symbolized just as well as true ones.

- 13.7. To say that everyone is good is tantamount to saying that there doesn't exist any person who is not good—it is *not* the case that there exists a person  $x$  who is not good, which is symbolized  $\sim\exists x\sim Gx$ .

We thus see that  $\forall$  is definable from  $\exists$  and  $\sim$ : For any property  $P$ , the expression  $\forall xPx$  is equivalent to  $\sim\exists x\sim Px$  (there doesn't exist an  $x$  that fails to have property  $P$ ).

- 13.8. In the other direction, if we start with  $\forall$  and  $\sim$ , we can get  $\exists$  thus: Given any property  $P$ , to say that there exists an  $x$  having property  $P$  is tantamount to saying that it is *not* the case that *every*  $x$  fails to have property  $P$ — $\sim\forall x\sim Px$ . In particular, to say that at least one person is good ( $\exists xGx$ ) is tantamount to saying that it is not the case that everyone is not good ( $\sim\forall x\sim Gx$ ).
- 13.9.  $\forall xPx\Rightarrow\exists xPx$  is one such formula. In a non-empty domain, if every element  $x$  in it has property  $P$ , then of course there is at least one  $x$  in it having property  $P$ , so the formula is certainly valid. But for an empty domain, the formula  $\forall xPx$  is true for any choice of  $P$ , as we have seen, but  $\exists xPx$  is certainly false, since there doesn't exist any  $x$  at all in an empty domain. Thus for any interpretation of  $P$  in an empty domain,  $\forall xPx$  is true and  $\exists xPx$  is false, so the entire formula  $\forall xPx\Rightarrow\exists xPx$  must be false.

- 13.10. Let  $F$  be the conjunction of the following three formulas:

$$F_1. \forall x\exists yRxy.$$

$$F_2. \sim\exists xRxx.$$

$$F_3. \forall x\forall y\forall z((Rxy\wedge Ryz)\Rightarrow Rxz).$$

That is,  $F$  is the formula  $(F_1\wedge F_2)\wedge F_3$ . Well,  $F$  is satisfiable in the domain of the natural numbers: Take  $Rxy$  to mean that  $x$  is *less than*  $y$ . Then

- (1) Of course, for any number  $x$ , there is a number  $y$  such that  $x$  is less than  $y$ .
- (2) No number  $x$  is less than itself.
- (3) For any numbers  $x$ ,  $y$ , and  $z$ , if  $x$  is less than  $y$  and  $y$  is less than  $z$ , then surely  $x$  is less than  $z$ .

Thus  $F$  is satisfiable in the infinite domain of natural numbers. Next we show that if  $A$  is any non-empty domain in which  $F$  can be satisfied, then  $A$  must contain infinitely many elements. Well, let  $R$  be any relation in  $A$  for which  $F$  is true. Since  $A$  is non-empty, it

contains at least one element, which we will denote  $a_1$ . By  $F_1$ , the element  $a_1$  must bear the relation  $R$  to some element  $a_2$ . Could  $a_2$  be the same element as  $a_1$ ? No; by  $F_2$ , it is false that  $Ra_1a_1$ . Thus  $a_2$  is distinct from  $a_1$ . Next,  $Ra_2a_3$  holds for some element  $a_3$  (by  $F_1$ ), and  $a_3$  must be distinct from  $a_2$  (again by  $F_2$ ). Also, since  $Ra_1a_2$  and  $Ra_2a_3$ , it follows that  $Ra_1a_3$  (by  $F_3$ ), and since  $Ra_1a_1$  cannot hold, it follows that  $a_3$  cannot be  $a_1$ . Thus  $a_3$  is distinct from  $a_1$ , also. Now we have three distinct elements  $a_1$ ,  $a_2$  and  $a_3$ . Next,  $Ra_3a_4$  holds for some element  $a_4$ , which by similar reasoning must be different from  $a_1$ ,  $a_2$  and  $a_3$ . Then  $Ra_4a_5$  holds for some new element  $a_5$ , and so forth. Thus we get an infinite sequence  $a_1, a_2, \dots, a_n, \dots$  of elements of the domain.

- 13.11. Yes, it is necessarily true. Here, of course, “anybody” means *somebody*, and we are to show that there is a person  $x$  such that if somebody drinks, then  $x$  drinks. Well, either it is true or false that somebody drinks. Suppose it is true. Then let  $x$  be any one of the people who drink. Thus  $Dx$  is true, hence of course  $\exists yDy \Rightarrow Dx$  is true (since a true proposition is implied by any proposition). On the other hand, if  $\exists yDy$  is false, then for *any* person  $x$ , it is true that  $\exists yDy \Rightarrow Dx$  (since a false proposition implies any proposition).

The formula  $\exists x(\exists yDy \Rightarrow Dx)$  is thus valid, and, as an exercise, the reader should prove it with a tableau.

- 13.12. Yes, it does logically follow! Since John loves Mary, John is a lover. Hence everyone loves John. Hence everyone is a lover. Hence, everyone loves everyone! In particular, Iago loves Othello.

Let  $Lxy$  mean “ $x$  loves  $y$ .” Then “ $x$  is a lover” is symbolically expressed as  $\exists yLxy$ . We have seen that if everyone loves a lover and if there exists so much as one lover, then everybody loves everybody! Well, “everyone loves a lover” is symbolized  $\forall x(\exists yLxy \Rightarrow \forall zLzx)$ . Also, “there exists a lover” is symbolized  $\exists x\exists yLxy$ . Of course, “everybody loves everybody” is symbolized  $\forall x\forall yLxy$ . Thus the formula  $\forall x\forall yLxy$  is a logical consequence of  $\forall x(\exists yLxy \Rightarrow \forall zLzx)$  and  $\exists x\exists yLxy$ . This can be proved by a tableau, and it is a good exercise for the reader.

- 13.13. Symbolically, the argument is this:

$$\frac{\begin{array}{l} \forall x\exists yMyx \\ \forall x\forall y(Myx \Rightarrow Pyx) \end{array}}{\therefore \forall x\exists yPyx}$$